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Predicting North Atlantic Temperature Trends with the Analogue Method using the MPI CMIP6 Grand Ensemble

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Lara Heyl

Abstract

Climate projections effectively estimate global surface temperature trends over a century, which are mainly dominated by global warming. In the North Atlantic, climate change impact on sea surface temperature (SST) has been comparatively modest and trends diverge significantly from the global trend. However, this region plays a crucial role in shaping the European climate, and decadal SST trend predictions of the North Atlantic can serve as outlook on the climate of the continent. To predict SST trends, the analogue method searches for analogues of observed SST patterns (reference) within a simulation ensemble. Due to ocean inertia, two analogous SST patterns evolve similarly on decadal timescales, enabling the analogue method to predict SST evolution. This is a simple and efficient alternative to more complex prediction methods that require new simulations. While the analogue method has been used for decadal predictions, its application to the prediction of ten-year SST trends remains unexplored. This thesis aims to extend the analogue method to predict ten-year trends of North Atlantic SST using the MPI CMIP6 grand ensemble of 30 members. We reconstruct the well-observed time series from 1960 to 2019 using the analogue method, and derive ten-year trends based on linear fits. Because the analogue method has only been applied for lead year predictions, we analyze its sensitivity on parameter choices from previous works. We introduce a new similarity definition that accounts for the time evolution of SST patterns and compare the resulting trend time series correlations with the results obtained by comparing time mean patterns. As the method cannot time external forcings like volcanoes, we assess the influence of volcanic eruptions on trend prediction skill. We report that the application of the analog method is robust for trend predictions with one simulation ensemble, the MPI CMIP6 grand ensemble. Volcanic influence can be accounted for by trimming the time series. Trend time series reconstruction exhibits a correlation with the reference of up to 0.9, and reveal an anticipated warming of the North Atlantic sea surface until 2029. The results suggest the effectiveness of the analogue method with a single simulation ensemble. A comparison of forecast skill shows that there is not one best parameter choice.

Zusammenfassung

Klimaprojektionen können Trends der mittleren globalen Oberflächentemperatur, die hauptsächlich durch die globale Erderwärmung beeinflusst wird, über ein Jahrhundert gut vorhersagen. Im Nordatlantik hingegen sind die Einflüsse des Klimawandels auf die Meeresoberflächentemperatur (*Sea Surface Temperature*, SST) vergleichsweise gering und die Temperaturtrends weichen erheblich vom globalen Trend ab. Diese Region spielt jedoch eine wichtige Rolle für das europäische Klima und dekadische SST-Trendvorhersagen im Nordatlantik können als ein Ausblick für das Klima des Kontinents dienen. Zur Vorhersage von SST-Trends sucht die Analog-Methode in einem Simulationsensemble nach ähnlichen Mustern zu beobachteten SST Mustern (Referenz). Aufgrund der Trägheit des Ozeans entwickeln sich zwei analoge SST-Muster auf dekadischen Zeitskalen ähnlich, sodass die Analog-Methode SST-Trends vorhersagen kann. Dies ist eine simple und effiziente Alternative zu komplexeren Vorhersagemethoden, die neue Simulationen erfordern. Die Analog-Method wurde schon für dekadische Vorhersagen benutzt, aber noch nicht für die Vorhersage von Zehn-Jahres-Trends. Diese Arbeit zielt darauf ab, die Analog-Method zu erweitern, sodass Zehn-Jahres-Trend von nordatlantischen SST unter der Verwendung des MPI CMIP6 Grand Ensembles von 30 Mitgliedern vorhergesagt werden können. Wir rekonstruieren den gut beobachteten Zeitraum von 1960 bis 2019 mit der Analog-Method und erstellen Zehn-Jahres-Trends mithilfe von linearer Regression. Da die Analog-Method bisher nur für Lead-Year Vorhersagen verwendet wurde, analysieren wir ihre Sensibilität auf verschiedene Parameter, die in vorherigen Arbeiten genutzt wurden. Wir führen eine neue Ähnlichkeits-Definition ein, die die zeitliche Entwicklung der SST-Muster berücksichtigt, und vergleichen die resultierenden Trend-Zeitreihen Korrelationen mit den Ergebnissen, die durch den Vergleich des zeitlichen Mittelwerts entstanden sind. Da die Methode externe Einflüsse wie Vulkane nicht vorhersagen kann, bewerten wir den Einfluss von Vulkanausbrüchen auf die Vorhersagefähigkeit von Trends. Wir berichten, dass die Anwendung der Analog-Method für Trendvorhersagen mit einem Simulationsensemble, das MPI CMIP6 Grand Ensemble, robust ist. Der Einfluss von Vulkanen kann durch Trimmen der Zeitreihen berücksichtigt werden. Die Rekonstruktion der Trend-Zeitreihen weist eine Korrelation mit der Referenz von bis zu 0,9 auf und sagt eine voraussichtliche Erwärmung der Meeresoberfläche des Nordatlantiks bis 2029 vorher. Die Ergebnisse deuten auf die Wirksamkeit der Analog-Method für ein einziges Simulationsensemble hin. Ein Vergleich der Vorhersagefähigkeit zeigt, dass es nicht die eine optimale Wahl der Parameter gibt.

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1 Introduction

Climate projections effectively estimate global surface temperature trends over the course of a century, but temperature changes in the next ten years directly influence people and environment and are therefore of immediate interest to societies. Global warming is leading to positive decadal trends for the global mean surface temperature while regions like the North Atlantic aren't as strongly affected. Decadal trends in these regions diverge from the global mean trends in the last sixty years. The drivers of change in the North Atlantic climate system are variations of the dominant multi-decadal features, such as the North Atlantic subpolar gyre (SPG) and the Atlantic Meridional Overturning Circulation (AMOC) (Griffies and Bryan [1997](#), Hermanson *et al.* [2014](#), Monerie *et al.* [2018](#), Yeager and Robson [2017](#)). These natural internal climate variations influence the North Atlantic on decadal scales to greater extent than anthropogenic forcings, which are the major driver of centennial climate changes. Because the North Atlantic sea surface temperatures (SST) influence the weather and climate over Europe, decadal trends of SST can serve as an outlook on the climate of the continent (Borchert *et al.* [2019](#), Hermanson *et al.* [2014](#), Sutton and Hodson [2005](#)).

Common predictions of future climates either take the form of the above mentioned climate projections of the next century, using different Shared Socioeconomic Pathways from the 6th Coupled Model Intercomparison Project (CMIP6) (Eyring *et al.* [2016](#)), or initialized predictions of the next decade (Boer *et al.* [2016](#)). Both approaches involve complex coupled climate models incorporating external forcings like greenhouse gases and anthropogenic and natural aerosols.

Simple and efficient methods for estimating near-term climate change impacts, particularly temperature trends, serve as valuable tools for skill benchmarks and comparison against more complicated prediction schemes. The analogue method proves useful even when large servers are not readily available for new simulations, utilizing existing information to generate new predictions.

Decadal predictions lie in a difficult regime between weather forecasts, characterized as an initial value problem, and long-term climate projections for the next century, governed by external forcings and represented as a forced boundary-value problem (Boer *et al.* [2016](#), Smith *et al.* [2013](#)). On decadal timescales, both the initial value and boundary-value problems are relevant, necessitating the integration of both internally generated and externally forced components to predict interannual to decadal climate

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evolution (Boer *et al.* 2016, Buckley *et al.* 2019).

While projections are a good estimation of long-term global climate change, they are imprecise on shorter timescales and for regional predictions. The dominant source of uncertainty in the first decade is internal variability of the climate system, preceding model and scenario uncertainties.

The concept of initialized predictions is to phase in the simulated with the observed internal variability by constraining internal variability in order to reduce uncertainty. But the initialization process comes with large computational expenses (Boer *et al.* 2016, Brown and Caldeira 2020) as well as the necessity to correct for initialization shock and model biases (Brown and Caldeira 2020, Smith *et al.* 2013).

The analogue method is a simpler way of applying the concept of phasing in internal variability by comparing observed and simulated climate states and finding the most similar match to the observed climate state in the simulations (Mahmood *et al.* 2021, Mahmood *et al.* 2022, Menary *et al.* 2021).

To predict SST trends, the analogue method searches for analogues of observed SST patterns within a data set. Because of ocean inertia, two SST patterns that are initially similar will remain close for some time in the future (Borchert *et al.* 2018, van den Dool 1994, Lorenz 1969), making it possible to predict future evolution of SST with the analogue method. The timescale depends on the initial similarity and the rate of error growth. An analogue of a present observed SST pattern can be used to predict the future of this pattern on the scale on which similar evolution can be assumed. The long memory of the North Atlantic suggests that this holds on decadal to multi-decadal timescales (Mahmood *et al.* 2021, Mahmood *et al.* 2022, Menary *et al.* 2021).

The analogue method is used in climate sciences and meteorology for simple prediction schemes. It can be performed using information from long historical records (van den Dool 1994) or existing simulation data sets, assuming the models providing these simulations are an adequate representation of reality (Ding *et al.* 2019, Menary *et al.* 2021). If the data set is long enough, a sufficiently analogous state can always be found (van den Dool 1994). Observational records are available only for a few decades (about 150 years), therefore, the probability of finding good analogues based on the natural analogue method is lower than for longer records. An alternative with longer data sets is to search analogues in existing simulation ensembles, i.e., analyzing a set of simulated data from an Earth system model (Ding *et al.* 2019, Mahmood *et al.* 2021, Menary *et al.* 2021, Zorita and von Storch 1999).

Earth system models (ESM) are designed to predict the most important features in the global climate, but their predictions are less precise on regional scales (Zorita and von Storch 1999). With the analogue method, analogues of an observed state can be defined globally (Mahmood *et al.* 2021, Mahmood *et al.* 2022) or regionally (Zorita *et al.* 1995, Zorita and von Storch 1999), making it possible to predict regional temperature

evolution with a simple method using the data produced with ESMs. Additionally, this procedure serves as a constraining principle so that analogues are in phase with the internal variability of the observed SST pattern (Mahmood *et al.* 2021, Mahmood *et al.* 2022, Menary *et al.* 2021). By selecting those simulation ensemble members that closely resemble the observed patterns of SST variability, the uncertainty arising from internal climate variability can be reduced. The analogue method can provide skillful predictions for ten or more years by combining information from historical simulations and climate projections (Mahmood *et al.* 2021). The SST patterns are compared for past states and can therefore only account for external, singular events (e.g. volcanic eruptions) when and if they imprint on the North Atlantic SST (Menary *et al.* 2021, Mahmood *et al.* 2021).

The aim of this thesis is to predict decadal SST trends with the analogue method. Previously, the method has been used only for lead year predictions of global and regional SST (Mahmood *et al.* 2021, Mahmood *et al.* 2022, Menary *et al.* 2021) for the time period 1960–2019. We introduce the expansion to decadal SST trend predictions in the North Atlantic using the Max Planck Institute 6th Coupled Model Intercomparison Project (MPI CMIP6, Olonscheck *et al.* 2023) grand ensemble generated with the the MPI Earth System Model (MPI-ESM, Mauritsen *et al.* 2019).

Given the novelty of this application, it is unclear whether the parameters utilized in previous studies, which were not specifically designed for trend predictions, contribute to skillful predictions in this context. Hence, this study analyzes the sensitivity of the analogue method’s ability to predict trends on different choices of parameters. Previous comparisons of SST patterns have predominantly relied on time mean states, possibly omitting time dependent information about pattern evolution. In this study, we introduce a novel time-dependent similarity definition in addition to established definitions (Mahmood *et al.* 2021, Menary *et al.* 2021), enabling a comprehensive assessment of trend prediction skill.

In pursuit of our research objective, we begin by evaluating the analogue method’s ability to predict trends through the reconstruction of the well-observed time series spanning 1960 to 2019. Ten-year trends are derived from linear fits and the correlation of this trend time series with observations is calculated. To apply the analogue method for trend predictions, we analyze its dependence on various parameters used in previous works. As the method cannot time external forcings like volcanoes, we briefly assess the influence of volcanic eruptions on trend prediction skill. For this purpose, we compare trend correlation under the influence of volcanic-driven upper ocean cooling with the correlation under exclusion of volcano years. We analyze the analogue method systematically on the dependence of the trend correlations on the main choices of parameters, time window length and number of selected analogues.

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Furthermore, we explore the impact of different similarity definitions, including a root-mean square error (RMSE)-based definition (Menary *et al.* [2021](#)), a correlation-based definition (Mahmood *et al.* [2021](#), Mahmood *et al.* [2022](#)), and a new time-dependent definition based on a 3D RMSE that considers the evolution of SST patterns. With this analysis, we can apply the analogue method to skillfully predict the trends for the next ten years, 2020–2029.

2 Data and Methods

2.1 Data

2.1.1 Observations

In this thesis, observations are used as a reference for the „real“ state of the Atlantic. The Hadley Centre Sea Ice and Sea Surface Temperature dataset (HadISST) by Rayner *et al.* (2003) is used and further on called *reference*. This dataset contains (SST) state estimates from monthly SST data based on available SST observations. The data is interpolated to a $1^\circ \times 1^\circ$ grid using an optimal interpolation procedure with reduced scale. The observations are available for the period from 1870 to 2019 and were derived from ship and buoy measurements and from satellite-borne high-resolution radiometers.

For the purpose of comparing the reference with simulation data, the data set is projected onto the native MPIOM GR15 grid (see Section 2.1.3). The annual mean SST anomalies over the North Atlantic for the period 1960–2019 are shown in Fig. 2.1. The anomalies are calculated relative to the climatology from 1960–1990. The absolute temperatures in the North Atlantic in this time period range from 8.47°C to 9.63°C , with a mean of 9.04°C and a standard deviation of $\sigma_T = 0.35^\circ\text{C}$. It can be seen that the North Atlantic has been warmer than the climatology since the early 1990s.

Since we want to focus on the trends of temperature, Fig. 2.2 shows the slopes of

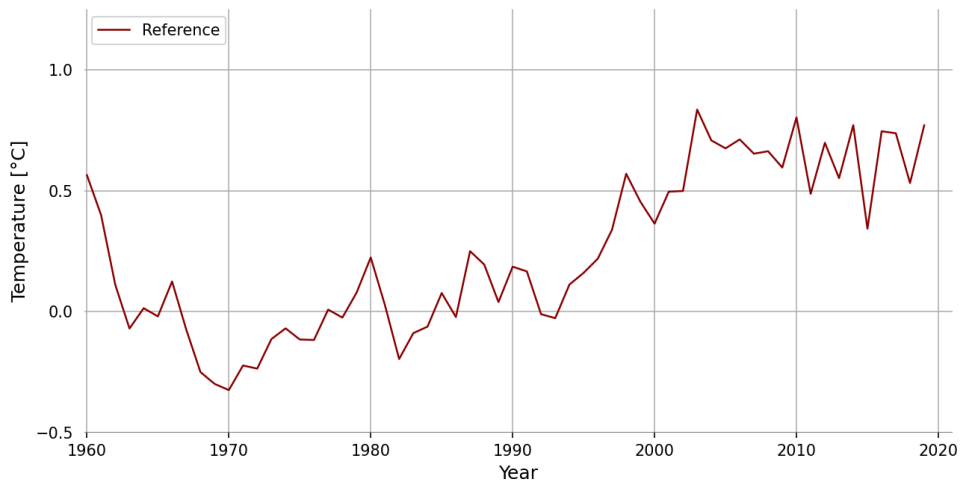


Fig. 2.1: Observed annual mean SST of the North Atlantic from 1960 to 2019, relative to 1960–1990 (HadISST, Rayner *et al.* 2003). Relative temperatures range from $T_{max} = 0.83^\circ\text{C}$, $T_{min} = -0.32^\circ\text{C}$, $\bar{T} = 0.25^\circ\text{C}$, $\sigma_t = 0.35^\circ\text{C}$

linear regression fits over moving ten-year intervals. The trends range from $T_{max} = 0.063^{\circ}\text{C}$ per year to $T_{min} = -0.077^{\circ}\text{C}$ per year, with a mean of 0.011°C per year and a standard deviation of $\sigma_T = 0.03^{\circ}\text{C}$ per year.

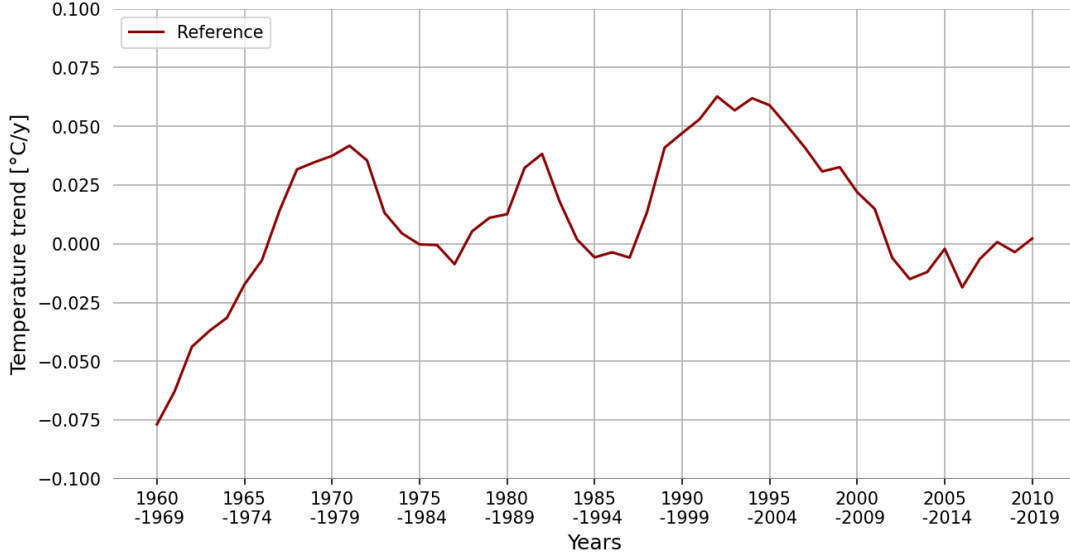


Fig. 2.2: Observed ten-year SST trends from 1960 to 2019 (Rayner *et al.* 2003). The trends range from 0.063°C per year to -0.077°C per year, with a mean of 0.011°C per year and a standard deviation of 0.03°C per year.

2.1.2 The Max Planck Institute Earth System Model (MPI-ESM)

2.1.2.1 Model description

In this work, we implement the analogue method for the Max Planck Institute 6th Coupled Model Intercomparison Project (CMIP6) grand ensemble, consisting of 30 members (Olonscheck *et al.* 2023). It is generated by the Max Planck Institute Earth System Model version 1.2 in low-resolution configuration (MPI-ESM1.2-LR; Mauritsen *et al.* 2019).

The MPI-ESM1.2-LR consists of four major components modeling the atmosphere, land, ocean and marine biogeochemistry, and a coupler (see Fig. 2.3). The atmosphere is represented by ECHAM6.3 and is directly coupled with the land model JSBACH3.2. The ocean is represented by the Max Planck Institute Ocean Model (MPIOM), coupled with marine biogeochemistry simulated by the Hamburg Ocean Carbon Cycle model, HAMOCC6. These main components are then coupled via the OASIS3-MCT coupler.

For the North Atlantic, the MPIOM is the most important part. The model represents the ocean globally as a primitive equation model on an Arakawa C-grid based on the hydrostatic and Boussinesq approximations. The low resolution configuration is designed in a bipolar orthogonal spherical coordinate system of 1.5° (native MPIOM GR15 grid; Jungclaus *et al.* 2013, Marsland *et al.* 2003) with one grid pole under Greenland and one under Antarctica. The effective horizontal grid cell lengths range from 15

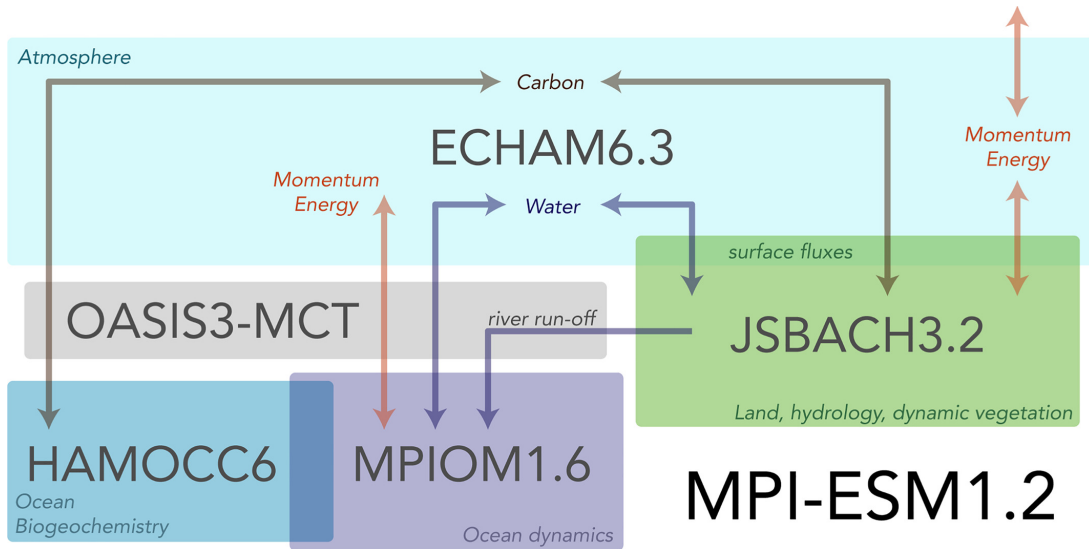


Fig. 2.3: Schematic overview of the major components of the MPI-ESM1.2. Arrows indicate interaction and energy transfer between components, overlap indicates direct coupling, while non-overlapping component blocks are coupled via the OASIS3-MCT coupler software (Plot taken from Mauritsen *et al.* 2019).

km around Greenland to 185 km in the tropical Pacific and are featured with 40 vertical levels of increasing thickness, with the uppermost layer ranging from 0 m to 12 m deep. Only the top layer, the surface of the ocean, is used in this work. This top layer interacts with the atmosphere, and with the land via river runoff. The coupling with OASIS3-MCT is implemented with a daily frequency (Mauritsen *et al.* 2019).

2.1.2.2 Simulation ensemble

The MPI CMIP6 grand ensemble is formed by branching from the MPI-ESM preindustrial control simulation (piControl; Wieners *et al.* 2023) at different states. The piControl simulation runs for 1000 years, with fixed forcings from a spin-up simulation in order to reproduce the quasi-equilibrium climate conditions from 1850 prior to the industrialization and the corresponding influence of anthropogenic forcings.

For the analogue method, we utilize the so-called historical simulations of the grand ensemble which span from 1850, branching off of various years of the piControl, to 2014 (Eyring *et al.* 2016). External forcings based on observations are applied to the simulated Earth system in order to recreate this time period. Forcings can be divided into natural and anthropogenic forcings. The former consist of the 11-year solar cycle and volcanic eruptions, while the latter include the effects of anthropogenic greenhouse gas, emissions, aerosols and global land use (Eyring *et al.* 2016).

By using this simulation data set, we utilize existing information to create new predictions. This requires less computational resources than creating new simulations for the purpose of decadal trend predictions, and it is faster, too.

2.1.2.3 Projections

The CMIP6 simulations provide a set of climate projections for the 21st century based on Shared Socioeconomic Pathways (SSP) related to Representative Concentration Pathways (RCP). These integrated assessment models (IAMs) are used for projections of future climate change (O'Neill *et al.* 2016).

The SSPs describe plausible trends of 21st century developments in society and ecosystems in the absence of further climate change or climate policies. The pathways assume different challenges to mitigation and adaption. The SSP2 scenario is called the medium path between SSP1, where sustainable developments proceed at a high rate, and SSP3, where moderate economic growth and high population growth are assumed (O'Neill *et al.* 2014).

RCPs describe possible future developments of anthropogenic drivers of climate change, such as land use and emissions of air pollutants and greenhouse gases (O'Neill *et al.* 2016). The RCPs are combined with SSPs to form IAMs (Kriegler *et al.* 2012, O'Neill *et al.* 2014) to produce climate projections that are relevant to societal concerns regarding climate change mitigation, adaption, and impacts.

In this thesis, we use the medium part of the range of future forcing pathways, the RCP4.5 which is a forcing pathway stabilizing at $4.5W/m^2$ at the end of the century. In combination with the SSP2 socioeconomic conditions, this scenario pathway is called SSP245. This scenario is used for decadal predictions, downscaling purposes and short-term predictions until 2030 (O'Neill *et al.* 2016) within the CMIP6. While we cannot know if the projections are correct for the future, we can assume that the SST patterns simulated by the projections are realistic patterns. Since the analogue method selects analogues based on their similarity with observations, we can use all realistic representations of SST patterns as a pool for the selection. We combine the SSP245 scenario projections starting from 2015 with the historical simulations ending in 2014, to make use of a longer time series of data.

2.1.3 North Atlantic region

Oceans have a large heat capacity and inertia, meaning that changes in temperature happen slower in the ocean than on land masses (Borchert *et al.* 2018). This makes variables like ocean heat content and sea surface temperature potentially predictable because the memory of the ocean, coming from the inertia, preconditions temperature predictability on these timescales in the North Atlantic (Borchert *et al.* 2018, Born *et al.* 2015, Buckley *et al.* 2019, Griffies and Bryan 1997, Pohlmann *et al.* 2009).

The North Atlantic SST is dominated by multi-decadal variations in SST like the North Atlantic subpolar gyre (SPG), the Atlantic Meridional Overturning Circulation (AMOC) and the North Atlantic Current. These large-scale SST patterns influence the European climate (Hermanson *et al.* 2014, Monerie *et al.* 2018, Sutton and Hodson 2005)

and are driving changes in the climate system while the ocean memory leads to predictability on decadal timescales (Borchert *et al.* 2018, Born *et al.* 2015, Buckley *et al.* 2019, Griffies and Bryan 1997, Pohlmann *et al.* 2009). This makes the North Atlantic SPG region interesting for decadal SST trend predictions, as the trends in the North Atlantic influence the air temperature relevant for European societies and the long memory leads to higher predictability.

For our application of the analogue method to trend predictions we choose the extratropical North Atlantic including the region around Greenland and the Nordic Seas (see Fig. 2.4). The corresponding box in terms of the native MPIOM GR15 grid is bounded by $i = (60, 200)$ and $j = (0, 100)$. This ensures the inclusion of the North Atlantic SPG south of Greenland and east of Labrador as well as the North Atlantic Current (Talley 2011).

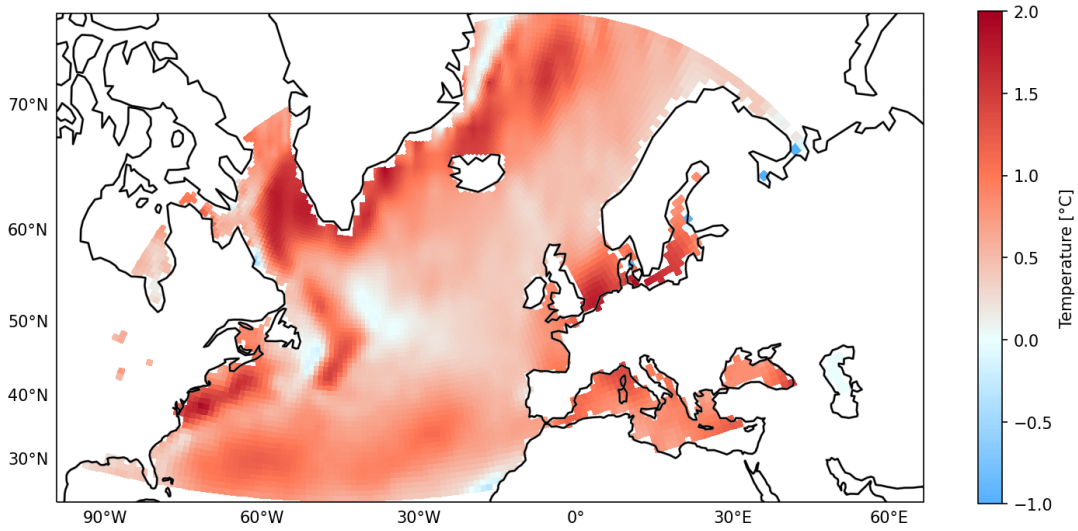


Fig. 2.4: Mean temperature anomalies observed in the North Atlantic in 2019, relative to the climatology 1960–1990 (Rayner *et al.* 2003). The model grid is projected onto a curvilinear grid for this plot, leading to a non-square box. The box in terms of the native MPIOM GR15 grid spans the indexes $i = (60, 200)$ and $j = (0, 100)$.

2.2 Methods

2.2.1 The Analogue Method

2.2.1.1 Description

The analogue method is based on a very simple concept: Predictions of the future can be made by considering the past. The simplest way of predicting the trend of the next ten years is to assume that the trend of the last ten years will continue (persistence). The reconstruction of the SST trend time series (Fig. 2.1) using this approach is shown in Fig. 2.5 and is consistent with shifting the trend time series by ten years. The correlation between the two trend time series is -0.08 , indicating no correlation or anti-correlation for different time intervals. The only time period that is predicted correctly for longer than one ten-year interval (intersections of reference and reconstruction) is in the ten-year intervals starting with 1981 until start year 1990. However, this is merely coincidental, and we have to assume from this reconstruction that a prediction for the next ten years (2020-2029) would not be skillful.

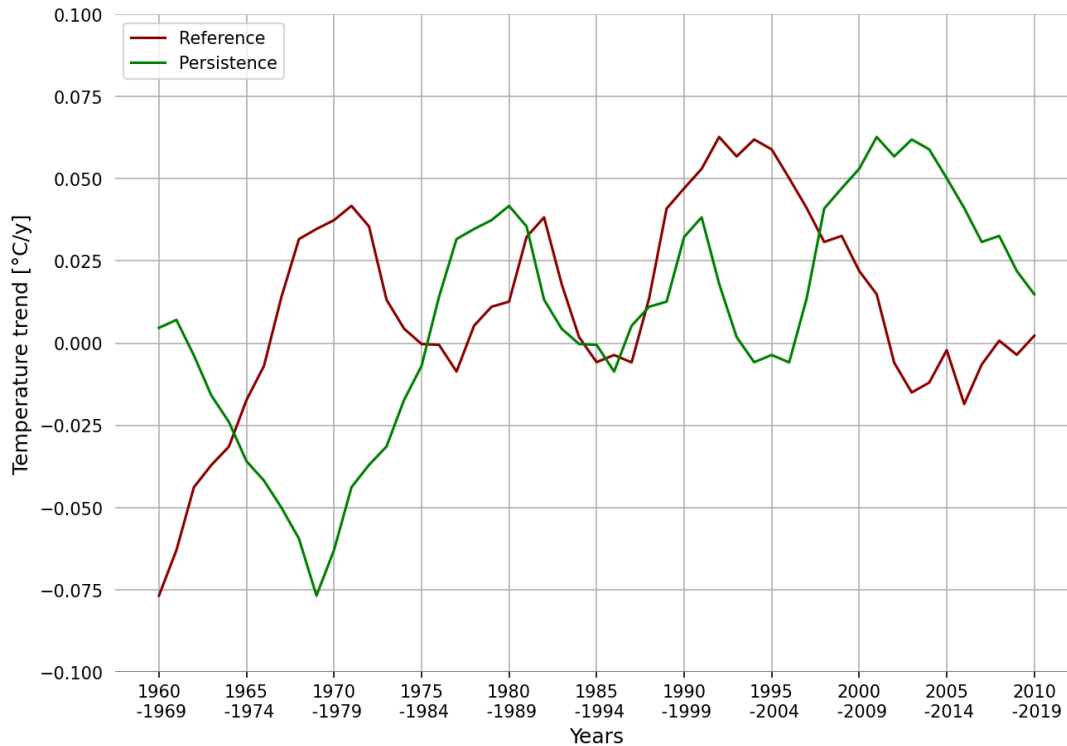


Fig. 2.5: Persistence trend time series, reconstructed (green) by assuming the trend of the last ten years. Observed trends are shown in red, both from Rayner *et al.* (2003). Correlation with the reference is $r = -0.08$.

In Fig. 2.6 we show a map of correlation of the persistence trend time series with the reference trend time series of each grid cell. We can see no correlation in the region of the North Atlantic SPG (south of Greenland), where we would expect skill because of the slowly changing multi-decadal variations. The higher correlation in the Mediter-

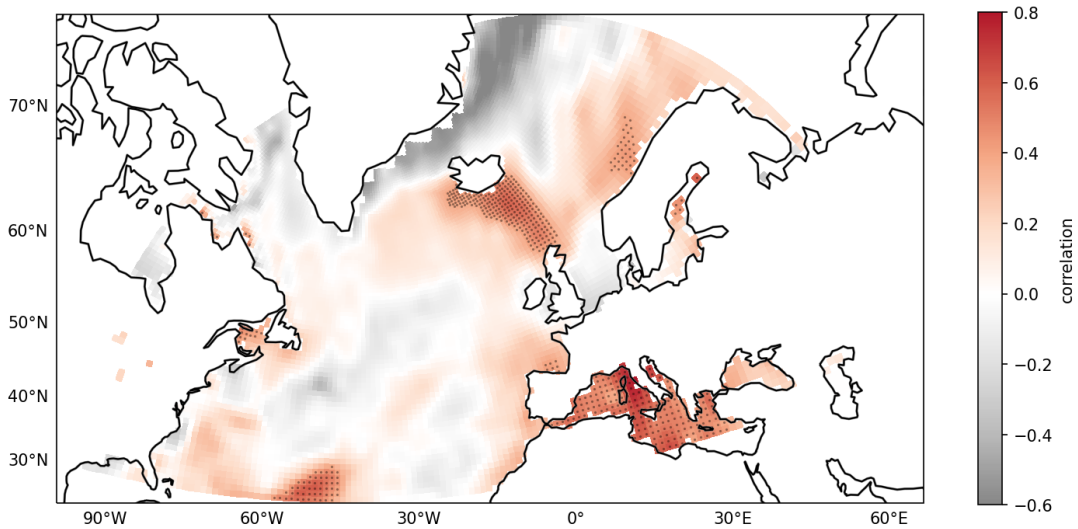


Fig. 2.6: Map of correlation between persistence 10-year trend time series and observed 10-year trend time series in each grid cell for 1960–2019 (Rayner *et al.* 2003). Stipples indicate values above 0.3 which are considered significant. The North Atlantic SPG regions shows no significant correlation with this approach.

anean Sea probably comes from a coincidentally different regional trend time series than the overall mean trend time series. However, this region is not the target region for the prediction.

In a more refined way of predicting the trend of the next ten years instead of assuming the trend will continue, we search for an analogous or similar SST pattern in the past and make use of the memory from the inertia of the ocean. We can assume that if two patterns of SST are similar at one point in time, they will evolve in a similar way for some time in the future, with the timescale depending on initial similarity and rate of error growth (van den Dool 1994, Lorenz 1969). Because of ocean memory, we can assume similar evolution over the course of ten years.

By selecting analogues for each time step from 1960–2019, we can reconstruct the trend time series (Fig. 2.2) and estimate the ability of this method to predict the past by calculating the correlation between reference and reconstruction. With this correlation value, we can assume the prediction skill of the analogue method for future trend predictions.

If the data set is long enough, a sufficiently analogous set can always be found (van den Dool 1994). However, observational data has only been available for about 150 years, and the accuracy of observations is lower for the earlier data (late 1800s). Therefore, analogues based on the natural analogue method are often less skillful than simulation-based analogue approaches (van den Dool 1994, Hawkins *et al.* 2011, Menary *et al.* 2021). This alternative searches for analogues in simulation data sets instead of the observational records. Simulation data often consists of ensembles, i.e., more than one simulation, spanning longer time periods, and the larger data sets increase the likelihood of finding good analogues (van den Dool 1994, Menary *et al.* 2021).

In this thesis, we use, for the first time, the simulations from the MPI CMIP6 grand ensemble (Olonscheck *et al.* 2023) in the search for analogues.

2.2.1.2 Assumptions

The simulation-based analogue method relies on a number of assumptions.

We have already named the first one in the introduction: Two climate fields that are similar to each other will remain close on a certain timescale (van den Dool 1994, Lorenz 1969). On decadal timescales this means that a well observed climate field is needed in order to compare observed and simulated fields. For the North Atlantic, observational availability, predictability and oceanic memory make SST patterns the perfect candidate (Boer *et al.* 2016, Griffies and Bryan 1997, Menary *et al.* 2021). The procedure of finding an analogue for an observed SST pattern serves as a constraining principle and phases in internal variability between reference and analogue. This means the large-scale features like the North Atlantic SPG (see Section 2.1.3) are in-phase and will evolve similarly for decadal timescales.

The second assumption is that the Earth system model producing the simulations is an adequate representation of reality (Brown and Caldeira 2020, Ding *et al.* 2019). This depends on the simulation used. The simulation based analogue method operates under the assumption that the mechanisms of variability linking past and future variability of SST are consistent across simulations and observations, and that the patterns of these relationships remain stationary in time (Menary *et al.* 2021). This means that an anomaly pattern of North Atlantic SST that leads to a certain large scale SST pattern is the same for different points in time. In Menary *et al.* (2021), where North Atlantic SPG SST was the forecast variable, the authors opted for the use of many different simulations from CMIP archives, to ensure that this assumption holds. Since our forecast variable is also North Atlantic SST this assumption is directly applicable. This work thus concentrates on the MPI CMIP6 grand ensemble consisting of 30 members produced by the MPI-ESM.

2.2.1.3 Procedure

A prediction for the next ten years (2020–2029) can be created by searching for an analogous SST pattern (analogue) of the mean SST patterns observed in a recent time window (e.g. 2010–2019; Mahmood *et al.* 2021, Menary *et al.* 2021). This means calculating the mean SST patterns over all possible time windows of the same length for all simulation ensemble members and comparing them with the observed pattern. The analogues are selected by finding the most similar matches to the observed pattern.

The result is a set of analogues for the recent past (in this case 2010-2019). The prediction is made by taking the data of the following ten years for each of these analogues. The analogues are chosen independently of their original time stamp and provide data

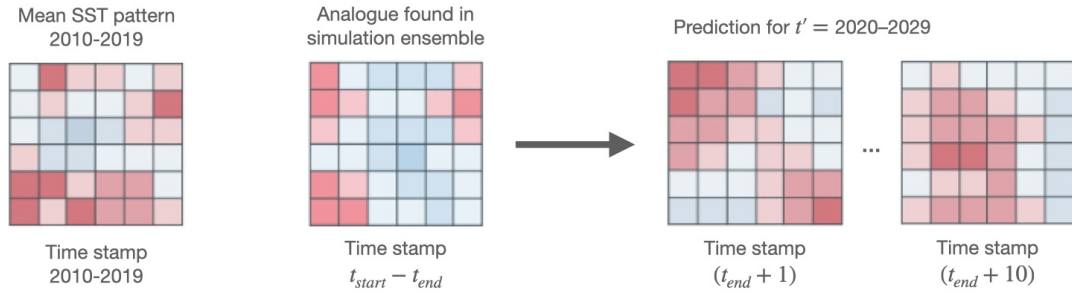


Fig. 2.7: Schematic overview of the analogue method prediction procedure. Squares depict SST patterns. The analogue found in the simulation ensemble has a different time stamp than the compared time window 2010–2019, and the data of the analogue goes beyond the compared time window, giving 10 years of annual SST data for the prediction of 2020–2029.

for at least ten years beyond the compared time window. Therefore, an independent prediction can be made using the evolution of the following ten years, assuming that similar SST patterns evolve similarly on these timescales (see Section 2.2.1.2).

A schematic depiction of this procedure is shown in Fig. 2.7. The analogue’s time stamp is removed and fixed to the time stamp of the reference, so that $t'_{start} = 2010$. While the comparison is calculated with time mean SST patterns over the time window length, the prediction consists of annual SST patterns. With the spatial mean SST of each prediction year, we can create a time series for the next ten years and obtain the ten-year trend (see Section 2.2.2).

2.2.2 Time series reconstruction

Prediction skill is measured by comparing the prediction made with a specific method with observations, and assessing the method’s ability to predict the evolution of the climate system (Boer *et al.* 2016). For future predictions, a comparison with observations is only possible once the future has become the present. The skill is therefore measured by comparing a reconstruction of a historical time series with observations in the same time period. The observations used as a reference in this thesis come from the HadISST (Rayner *et al.* 2003) data set (see Section 2.1.1).

Fig. 2.2 shows the trend time series of the reference in the ten-year intervals 1960–1969 to 2010–2019. We refer to these ten-year intervals by their start year, t'_{start} , e.g., 1960 to 2010. We want to reconstruct this trend time series with the analogue method in order to estimate the trend prediction skill. For each time step of the reference time series, we compare the previous time window of the reference with all time windows of the same length in the simulation data. For time windows of 10 years and the ten-year interval 1960–1969, this means comparing the time mean SST pattern observed in 1950–1959 with all 10-year time mean SST patterns calculated from the simulation ensemble. With the available simulation data from 1850–2100, we can create 231 such time mean patterns for each of the 30 ensemble members. The analogues are found by

selecting the most similar patterns by means of a similarity definition, and their time stamps t'_{start} and t'_{end} are set to 1950–1959.

The reconstruction of the trend is created by using the ten-year interval following the compared time period, in this case $t'_{start} = 1960$ to $t'_{end} = 1969$, of each analogue. The trend is calculated in the same way as in Section 2.1.1 by taking the slope of a linear regression fit for the ten-year spatial mean SST time series. Because SST changes slowly over a decade, a more complicated fit does not lead to a better trend estimation, and linear fits are sufficient. By proceeding in this way for every ten-year interval of the reference trend time series, we can reconstruct the entire time series.

For comparison, lead year prediction would aim at the reconstruction of the individual years from 1960–2019. Each analogue gives ten years of reconstruction and the lead year refers to the index of the year in this ten-year interval, e.g., for the analogue of 1950–1959, the reconstruction uses the following ten years, 1960–1969. Lead year one for this first interval is 1960, lead year 2 1961, and so on. Lead year predictions are usually presented as a mean, commonly lead year 2–5 combined (Boer *et al.* 2016. Menary *et al.* (2021) used lead year 2–10 combined, meaning the mean of lead year 2 to lead year 10 as a time series. Trend prediction instead use the entire time series available and calculate a slope from the linear fit.

Additionally to reconstruction, the analogue method can provide predictions by comparing all time mean SST patterns up to 2019, giving the trends until 2029 created from the analogues.

2.2.3 Choice of parameters

The trend time series reconstruction described in Section 2.2.2 requires three choices that we have to decide on. The first appears when preparing the comparison, the time window length p . The second when deciding how many analogues to select, the analogue ensemble size K . Lastly, we have to decide on a definition of similarity by which we choose the "best matches".

2.2.3.1 Time window length and analogue ensemble size

The first choice is the time window length p over which the time mean SST patterns are calculated and then compared. And secondly, the number of selected analogues, the analogue ensemble size K , has to be defined.

Previous studies have done brief analyses of the time window length p , suitable for lead year predictions. Good results were obtained for 20-year time windows (Mahmood *et al.* 2021) as well as 6 and 9 year windows (Mahmood *et al.* 2022). Menary *et al.* (2021) deemed the time window of 35-years most suitable for lead year 2–10 combined. In this thesis, we want to apply the analogue method to trend predictions, therefore the

previously used choices might not be the best for this application. We analyze the skill of trend predictions for time window lengths from 1 year up to 40 years to find the most suitable choice. Additionally, we can use this analysis for a sensitivity test.

The analogue ensemble size is limited by the number of independent members. While we can allow multiple analogues from the same member with non-overlapping time periods (Menary *et al.* 2021), shorter time windows would lead to more analogues from one member than longer time windows. For consistency with different time windows, we add the constriction that only one analogue per member and time window is allowed. Therefore, we choose between 1 and 30 different analogues per time step. Previous studies selected 10 and 20 (Mahmood *et al.* 2021) as well as 30 and 50 (Mahmood *et al.* 2022) best matches from a pool of 40 and 212 members, respectively. In Menary *et al.* (2021), a total of 100 analogues was selected from the CMIP5 and CMIP6 archives.

The sensitivity of the analogue method's ability to predict trends is tested for these two parameter choices.

2.2.3.2 Similarity definition

To find the most similar SST pattern for each reference time step, we have to apply a similarity definition in order to identify analogues. The similarity definition needs to be well suited for the analogue variable and spatiality of the problem. We want to predict North Atlantic sea surface temperatures with two-dimensional input data (SST patterns). Previous studies have recommended using spatial root-mean-square-error (2D RMSE, (Eq. 2.1), Menary *et al.* 2021) and spatial pattern correlation or Pearson correlation (2D correlation, (Eq. 2.2), Mahmood *et al.* 2021, Mahmood *et al.* 2022). We compare these two definitions of similarity to analyze the sensitivity of the analogue method to the mathematical comparison metric.

$$d(\bar{T}_{ref}(t_m), \bar{T}_s(t_n)) = \sqrt{\frac{\sum_{ij} (\bar{T}_{i,j,s}(t_n) - \bar{T}_{i,j,ref}(t_m))^2}{N}} \quad (2.1)$$

$$d(\bar{T}_{ref}(t_m), \bar{T}_s(t_n)) = \frac{\sum_{i=1} (\bar{T}_{i,j,s}(t_n) - \bar{T}_s(t_n)) (\bar{T}_{i,j,ref}(t_m) - \bar{T}_{ref}(t_m))}{\sqrt{\sum_{ij} (\bar{T}_{i,j,s}(t_n) - \bar{T}_s(t_n))^2 \sum_{ij} (\bar{T}_{i,j,ref}(t_m) - \bar{T}_{ref}(t_m))^2}} \quad (2.2)$$

Where i, j are the grid cell indices, N is the total number of grid cells, s is the simulation ensemble member number, $t_{n,m}$ is the start year of the time mean window of reference and simulation, $\bar{T}$ is the time mean SST matrix and $\bar{T}(t_n)$ is the mean over all grid cells.

Independently of the specific similarity metric, these definitions rely on the time mean state of the compared variable. The similarity is calculated between the time mean SST patterns over a certain time window p (Mahmood *et al.* 2021, Mahmood *et*

al. 2022, Menary *et al.* 2021). This means that any time dependent features within the time window do not appear in the comparison and analogues are not selected based on their similarity over time. To account for this time evolution of SST patterns, we introduce a new definition of similarity with an additional time dimension. With this new similarity definition we compare each annual SST pattern in the time window, creating a timeline of similarity for each time step, where $year_1$ of the reference time window is compared with $year_1$ of the respective simulation member time window.

We base this new definition on the 2D RMSE-based definition (Menary *et al.* 2021) by adding a sum over the time index t (3D RMSE, (Eq. 2.3)):

$$d(T_{ref}, T_s) = \sqrt{\frac{\sum_{ijt} (T_{i,j,t_n,s}) - T_{i,j,t_m,ref})^2}{len(t) \cdot N}} \quad (2.3)$$

Where $t_{n,m}$ is the additional time index referring to the specific year. Both run over the same length of years, $1 \dots p$, but their yearly time stamps are referring to the $t_{n,m}$.

For each of the three similarity definition we conduct an analysis of the previously named parameters (Section 2.2.3.1), creating 1200 parameter sets (K, p) each. With every parameter set, the trend time series of 1960–2019 is reconstructed, with a prediction until 2029. The correlation of the overlapping time period (start years 1960 to 2010) with the reference gives a measure for the prediction skill of this parameter set and similarity definition. Additionally, the use of different similarity definitions opens the opportunity of creating prediction ensembles that provide a measure of prediction uncertainty. With this in-depth analysis, we can test the analogue method on robustness for trend predictions.

2.3 Volcanic eruptions

Major volcanic eruptions can influence the temperature of oceans and atmosphere. Volcanoes produce sulfate aerosols that lead to back reflection of incoming shortwave radiation for about two years after a volcano eruption (Robock 2000, Driscoll *et al.* 2012). This affects the Earth’s climate by cooling the surface when a sufficient amount of sulfur containing gases enter into the stratosphere. The volcanically induced surface cooling results in a short-lived cooling of the upper ocean (Swingedouw *et al.* 2015) on the timescale of the e-folding lifetime of the sulfate particles of about 12–14 months (Robock and Mao 1995, Driscoll *et al.* 2012). Because we use annual data and do not assess when in a year a volcano erupted, we assume volcanic influence for the year of the eruption plus the two following years, in accordance with literature.

Three volcanoes have affected the North Atlantic since 1960: Agung in 1963, Chichon in 1982 and Pinatubo in 1991. This means that we can expect repercussions in the North

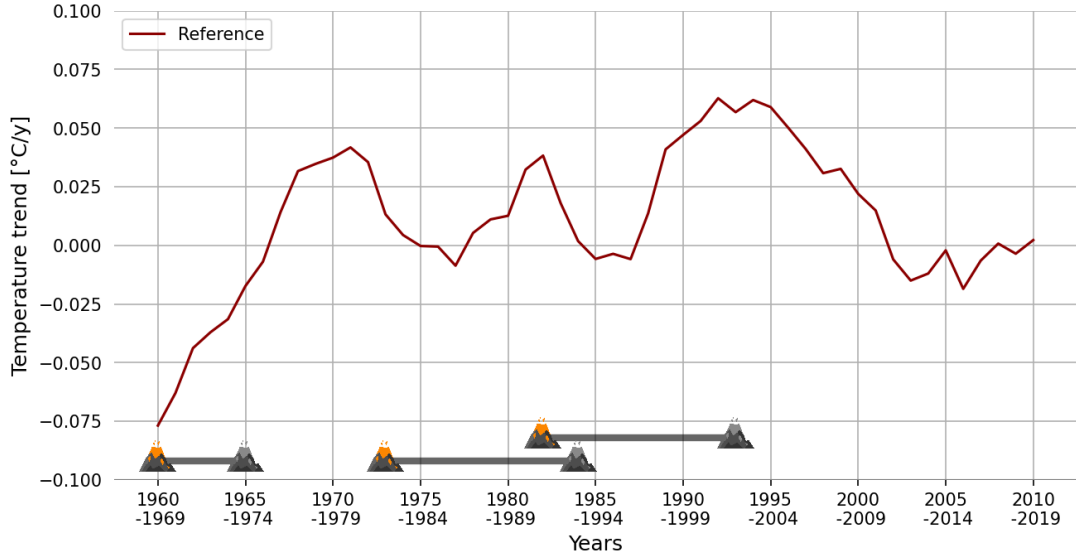


Fig. 2.8: Observed 10-year trend time series of annual SST anomalies in the North Atlantic (Rayner *et al.* 2003). The x-axis shows the ten-year intervals for the trend. Volcanic influence is highlighted. The colored volcano indicates the ten-year interval where the eruption first appears (1963, 1982, 1991), while the grey volcano shows the end of the influence. The grey line spans the whole time period in which the volcano influences the trend. The influences from Chichon (1981) and Pinatubo (1991) overlap.

Atlantic for the following years:

Volcano years: 1963, 1964, 1965 and 1982, 1983, 1984 and 1991, 1992, 1993

In this work, we expect an influence of the volcanic-driven upper ocean cooling on the ten-year trends (Timmreck *et al.* 2016). Every ten-year interval including one or more volcano years is expected to have a smaller or negative trend than if the same time period did not experience any volcanic eruptions. The analogue method compares SST patterns for the past, so any volcanic eruptions happening in this time period can be accounted for. However, the volcano years cannot be predicted by the analogue method by design (Mahmood *et al.* 2021, Menary *et al.* 2021). Today, there is no reliable method for predicting volcanic eruptions on decadal timescales so that we cannot prescribe any volcanoes for the prediction of the future, e.g., 2020–2029. Previous studies on the analogue method have not researched this topic further than mentioning that the analogue method cannot predict volcanoes (Mahmood *et al.* 2021, Menary *et al.* 2021).

To realistically assess the skill of the analogue method for future trend predictions, we compare the trend time series correlation with and without volcano years included in order to see if the volcanoes influence the prediction skill. If the correlations differ significantly, exclusion of all ten-year intervals with volcano years is advisable. This means that the trend time series with start years of the ten-year intervals from 1960–2010 is trimmed to contain only intervals starting with the following years:

[1966, 1967, 1968, 1969, 1970, 1971, 1972, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010]

In Fig. 2.8 we have highlighted the time intervals where volcanic eruptions have influenced the North Atlantic. Large parts of the time series are impacted by these events. When we exclude these time intervals, we obtain a shorter time series but can calculate a correlation nonetheless, containing all trend values for the recent past following 1994. Exclusion of volcano years from the correlation is the simplest way of comparing the influence of volcanoes because no simulation data without volcanic forcing is available for the MPI CMIP6 grand ensemble.

In general, the role of external forcings like volcanic eruptions in decadal prediction skill assessments remains a question unanswered. Climate models often fail to accurately simulate the dynamics after a volcanic eruption (Driscoll *et al.* 2012). Time-varying radiative forcings from greenhouse gases, anthropogenic and natural aerosols, including volcanoes, are a source of skill in the North Atlantic (Yeager and Robson 2017) and while we do not provide an in-depth analysis of this, the exclusion of volcanoes from the skill assessment in this thesis can be a starting point for further analysis.

2.4 Analysis

The data analysis in this thesis is done with the Climate Data Operator (CDO) software (Schulzweida 2023), for preparation of the data, and with the programming language Python, using the integrated Jupyterhub on the German Climate Computing Center (Deutsches Klimarechenzentrum DKRZ) high-performance computing system Levante, for all other calculations and plots.

2.4.1 Bootstrapping

In order to get a statistical significance estimation for the selected analogues and the trend time series, we choose a bootstrapping procedure. Bootstrapping is a resampling procedure for one-sample settings. The idea is to treat a finite sample as similarly as possible to the unknown distribution from which it was drawn. This statistic can be applied to any test statistic (Wilks 2011).

The bootstrapping procedure is applied to generate a significance threshold for the trend correlations. In previous papers, the 90th (Menary *et al.* 2021) and 95th percentile (Mahmood *et al.* 2021) have been suggested. By analyzing both tails of a 1000-member bootstrapped probability distribution of the resampled trend time series, we calculate the significance of correlations at the 95% level. This is done for each similarity definition and parameter set (K, p) , individually.

We use this method for the selection of analogues as well by sub-selecting random members for each time step. This aims at distinguishing the significance of results from a randomly chosen distribution. The bootstrapping of the selection mechanism is very computer-intensive which is why it is done with a lower number of samplings, 250

per analogue ensemble size K and time window length p . In total $250 \times 1200 \times K \times 61$ years samples are created for all three similarity definitions. First, for every reference year and each p, K , K members are randomly selected from the 30 ensemble members of the MPI-ESM. This replaces the selection by the similarity definition. Afterwards, the time series 1960–2020 is reconstructed and ten-year trends are calculated.

3 Results

3.1 Trend time series reconstruction with the analogue method

3.1.1 Trend time series reconstruction

This thesis focuses on the ten-year trend predictions generated with the analogue method for the MPI CMIP6 grand ensemble comprising 30 members. We start with the parameters of previous studies applied to lead year predictions in the North Atlantic to see if they are applicable to trend predictions. Fig. 3.1 shows the ten-year trend time series reconstruction of 1960–2019 by selecting the 10 best ($K = 10$; Mahmood *et al.* 2021) analogues found when matching 35 year time mean SST patterns ($p = 35$; Menary *et al.* 2021) of reference and simulations with a 2D RMSE similarity definition (Menary *et al.* 2021). For skill estimation, we calculate the Pearson correlation coefficient during the overlapping time period (1960–1969 to 2010–2019). The reconstruction in Fig. 3.1 has a correlation with the reference of $r = 0.64$. The trend bootstrapping significance at the 95% level (see Section 2.4.1) is 0.37, so the reconstructed trend is above the threshold and significant. Therefore, the application to trend reconstruction with the parameters suggested by studies about lead year reconstruction with the analogue method is successful.

We want to compare the ability to reconstruct the trend time series with the persistence shown in Fig. 2.5, where we assume the trend of the previous ten years will continue for the next ten years. The trend time series reconstruction with persistence has a correlation of -0.08 , meaning no correlation or partly anti-correlation. The correlation of the reconstruction with the analogue method of 0.64 is significantly higher.

The reconstructed trend time series aligns with the reference for a few time intervals, such as the ten-year intervals with start year 1965 to start year 1972, but is almost anti-correlated for other intervals, such as start year 1960 to 1965 or 1972 to 1980 and 1982 to 1989. Notably, these intervals coincide with negative or decreasing trends of the reference time series, and all these ten-year intervals include one of the years 1963, 1982 or 1991. These years are when major volcanic eruptions resulted in an upper

3.1 Trend time series reconstruction with the analogue method

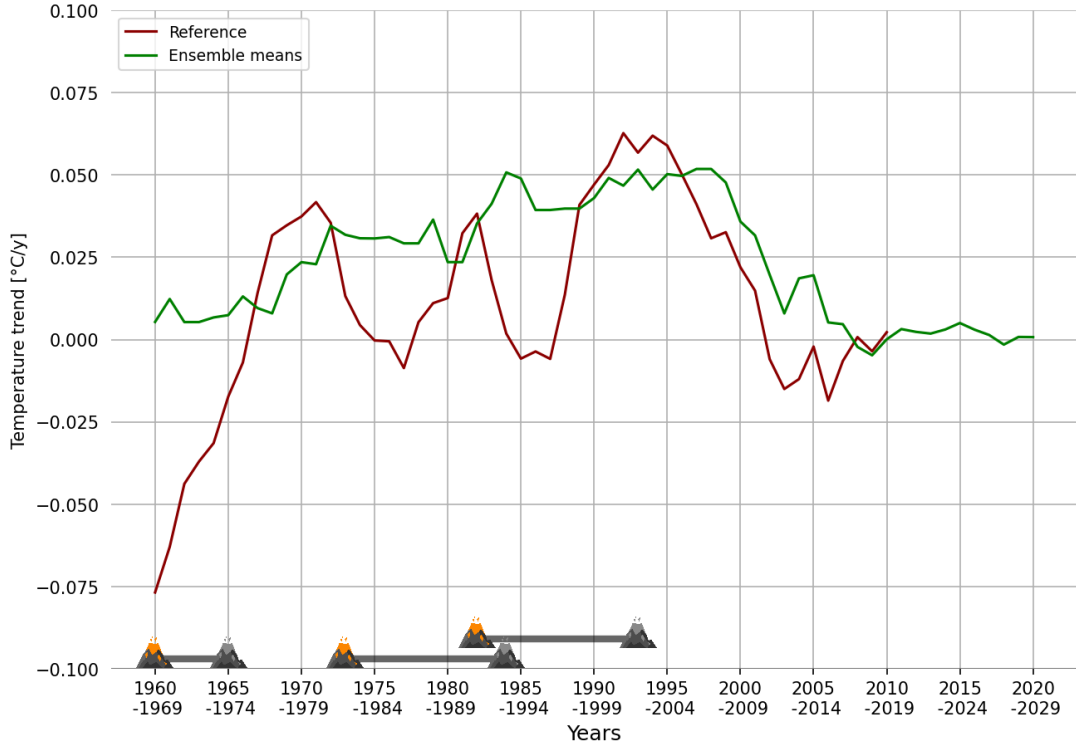


Fig. 3.1: The trend time series of the reference and one example analogue reconstruction done with the parameters and 2D RMSE-based similarity definition suggested by Menary *et al.* (2021), $p = 35$, and Mahmood *et al.* (2022), $K=10$. The x-axis denotes ten-year intervals, the first of which is indicated by start year 1960. The y-axis represents the slope of the trend fit in $^{\circ}\text{C}/\text{y}$. The red line shows the observed trend time series of the reference (Rayner *et al.* 2003), while the green line denotes the analogue ensemble mean of the reconstruction. Triangles indicate a volcanic eruption and the grey lines span the influence of this eruption. In three years (1963, 1982, 1991) a major volcanic eruption influenced the North Atlantic. These external events cannot be predicted by the analogue method. The correlation between reference and reconstruction is $r = 0.64$.

ocean cooling in the North Atlantic. Every ten-year interval from start year 1960 to 1963, 1973 to 1982 and 1982 to 1991 includes years marked by volcanic eruptions. As the aerosols responsible for upper ocean cooling persist in the atmosphere for approximately two years, the two years following an eruption are considered to be influenced by the volcano as well (see Section 2.3). Consequently, trends over ten-year intervals for start years from 1960 to 1965 and 1973 to 1993 (highlighted by grey lines in Fig. 3.1) are impacted by externally forced cooling that cannot be predicted with the analogue method.

To assess the impact of volcanic eruptions on our reconstruction skill, we compare the correlation of the entire overlapping trend time series with the correlation after removing all ten-year trends in the highlighted time intervals. The correlation, excluding volcano years, stands at $r = 0.79$. Upon examining the remaining time series, we observe a much closer alignment between the reference and reconstruction. When including volcano years, we expect the method to predict something beyond its capability. Additionally, this inclusion may lead to underestimating the ability to predict

time periods without volcanic influence. Considering predictions of the future where volcanic influence is unknown, evaluating prediction skill while excluding volcano years provides a closer approximation to the skill we can expect for scenarios where no volcanic eruptions induce upper ocean cooling. For the remainder of the study, we focus on correlations without volcano years.

3.1.2 Map of trend skill

The trend is calculated from the time series of annual North Atlantic spatial mean of SST derived from the analogues. We now want to test the assumption that prediction skill in the North Atlantic can be expected because of the dominant influence of internal variability by the North Atlantic SPG. For this purpose, we show a map of trend prediction skill in Fig. 3.2, created for the same parameters $K = 10$, $p = 35$ (Fig. 3.1) by comparing the reference trend time series of each grid cell. The map reveals significantly high skill in the SPG region south of Greenland and between Greenland and Iceland. The areas with lower skill, depicted in grey, are in regions where models typically struggle with predictions due to river runoff and ocean-atmosphere interaction east of North America, and sea ice influence south of the pole. The larger grey area east of North America comes from the turbulent boundary between subpolar and subtropical gyre. These factors happen on shorter time scales than large-scale ocean currents, resulting in much shorter memory, and the skill on decadal timescales is lower compared to the multi-decadal SPG.

An improvement in prediction skill is evident when comparing this trend skill map

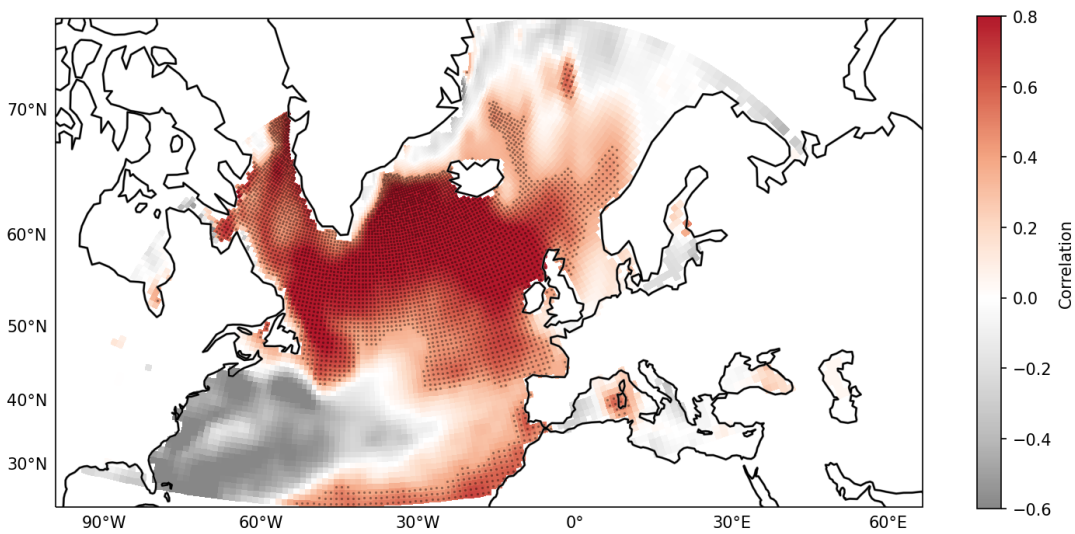


Fig. 3.2: Map of trend prediction skill for $K = 10$, $p = 35$ without volcano years in the region of the North Atlantic. Stipples indicate significance above the 95% level. The simulation grid is different from the spherical coordinates, therefore the box is not squared in this map projection. The significance level lies at 0.37. High skill is found in the region of the North Atlantic SPG, as expected.

with the reference persistence map in Fig. 2.6. The persistence-based reconstruction demonstrated few areas of significant skill, and none in the SPG region. Using the trend reconstructed with the analogue method improves the prediction skill compared to simple persistence.

The analogue method compares large-scale SST patterns in the North Atlantic. The North Atlantic SPG is a dominant multi-decadal feature that is generally well simulated by MPI-ESM. Our findings indicate enhanced skill in regions characterized by dominant SST patterns when predicting trends with the analogue method.

3.2 Comparison of parameters and similarity definitions

3.2.1 Overview of different similarity definitions

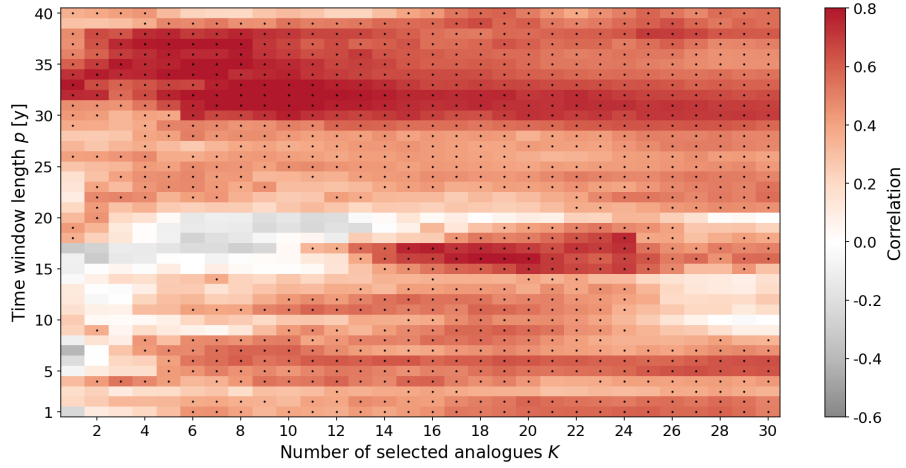
The previous section aimed at the implementation of the analogue method for trend predictions using the same parameters used previously for lead year predictions. Now, we want to analyze this further and conduct a sensitivity analysis of the analogue method concerning the parameters time window length, p , and number of selected analogues, K , as well as the similarity definition. The optimal parameters can be different for trend predictions than those for lead year predictions. We vary the time window length from 1 year to 40 years and select up to 30 different analogues per reference time step.

For each parameter set (K, p) , we create a reconstructed trend time series by selecting analogues with the 2D RMSE-based definition for the time series 1960–2020, creating a trend time series like the one in Fig. 3.1. As an overview, the correlation coefficients with the reference time series are depicted in Fig. 3.3 (a) as a function of (K, p) . For each set, a separate trend bootstrapping is performed, giving individual 95% significance levels with a mean of 0.36. The correlations are distributed with a range from $r_{max} = 0.91$ for $K = 8, p = 31$ to $r_{min} = -0.41$ for $K = 1, p = 7$. Notably, this 2D RMSE-based analogue method exhibits sensitivity to the time window length p , while sensitivity to K only appears in certain areas of p , particularly around $p = 16$ years. Highest skill can be found for all K in the range of $p \in \{31, \dots, 38\}$ years, and for $K \in \{14, \dots, 24\}$ for $p = 15, 16$ and 17 years.

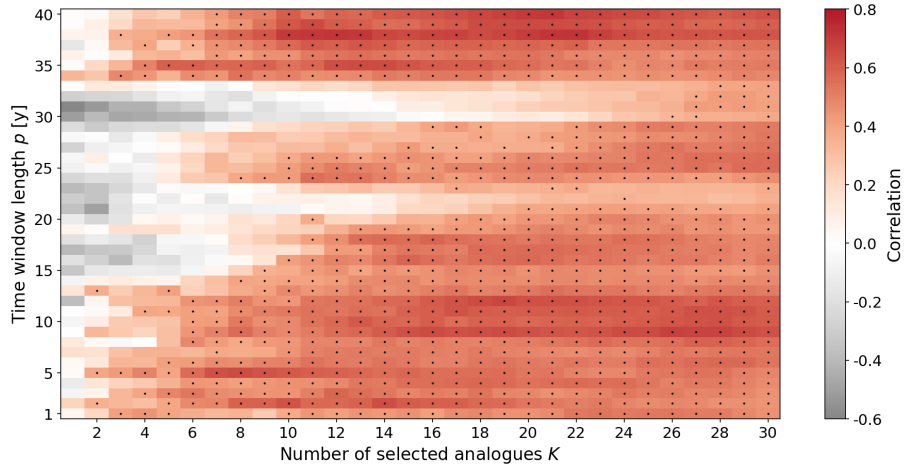
For comparison, we implement the 2D correlation-based definition found in Mahmood *et al.* (2021), which is shown in Fig. 3.3 (b). The correlations are distributed with $r_{max} = 0.72$ for $K = 22, p = 38$ and $r_{min} = -0.61$ for $K = 1, p = 31$, with a mean 95% significance level of 0.36. For this similarity definition, we can see sensitivity to the number of selected analogues, where smaller K values lead to less skill compared to $K > 3 \dots 5$. For larger K , the sensitivity to the time window length p is largely mitigated, with a marginally less skillful area around $p = 23$ and $p = 30$ years.

Lastly, we introduce a new similarity definition based on time dependent RMSE (3D

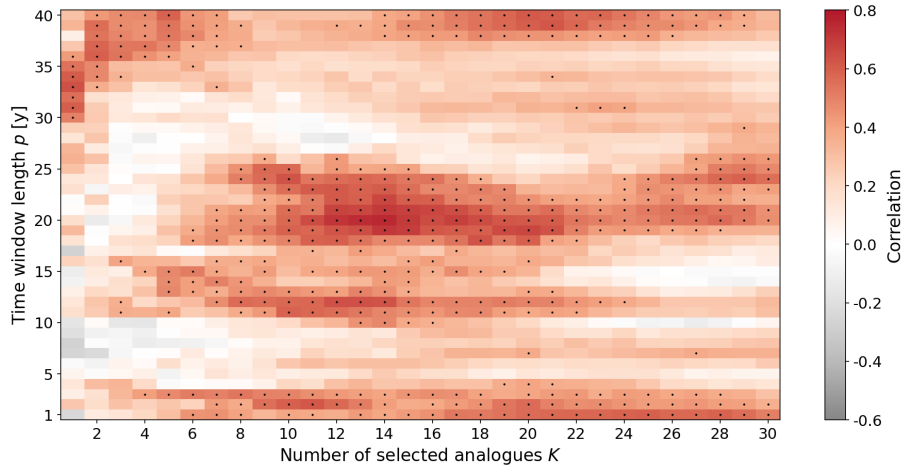
3 Results



(a) 2D RMSE-based definition. Correlations are distributed with $r_{max} = 0.91$, $r_{min} = -0.41$.



(b) 2D correlation-based definition. Correlations are distributed with $r_{max} = 0.72$, $r_{min} = -0.61$.



(c) 3D RMSE-based definition. Correlations are distributed with $r_{max} = 0.76$, $r_{min} = -0.27$.

Fig. 3.3: Trend correlation of the reconstructed trend time series with the reference (Rayner *et al.* 2003) for different parameter sets p, K . a) 2D RMSE, b) 2D correlation, c) 3D RMSE-based definition. The number of selected analogues (K) is represented on the x-axis, and time window length (p) shown on the y-axis. The color bar indicates the correlation, where darker red signifies a higher correlation. Stipples indicate correlations above the 95% significance level from the trend bootstrapping for each individual parameter.

RMSE, see 2.2.3.2). This approach aims at leveraging information hidden by taking the average over a time window. Especially for longer time windows, this is expected to exhibit different behavior than the two time mean-based definitions. The results are shown in Fig. 3.3 (c), with $r_{max} = 0.76$ for $K = 14$, $p = 20$ and $r_{min} = -0.27$ for $K = 1, p = 9$. 95% significance level is at 0.36 on average. Here, a distinct area of highest skill emerges at $K \in \{10, \dots, 20\}$ and $p \in \{18, \dots, 25\}$ years. While several parameter sets exhibit skill smaller than the 95% significance level, areas of high skill can be found.

For all parameter sets and similarity definitions, a separate bootstrapping is performed by selecting 250 random ensemble members for every time step. This ensures that the analogue method results are distinguishable from a random selection (see Section 2.4.1). For all three similarity definitions, the mean significance at the 95% level is 0.37, therefore being in the same range as the individual the trend time series bootstrapping shown in Fig. 3.3.

Overall, selecting $K = 1$ seems to be a suboptimal choice across all similarity definitions, with longer p yielding better results. The optimal parameter sets vary depending on the similarity definition employed, but consistently, we can identify parameter sets demonstrating high trend prediction skill. This indicates that the analogue method consistently yields significantly correlated SST trend reconstructions, with correlations up to $r = 0.9$.

3.2.2 Trend skill sensitivity to time window length

In this section, we continue the sensitivity analysis for the time window length because the correlation of the reconstruction of trends depends the most on this parameter. We have seen that all three similarity definitions show less sensitivity to the number of selected analogues K , especially for the time mean-based definitions. The dependence on time window length p shows in stripes and areas of significant values in the Fig. 3.3 (a)-(c). For further analysis, we compare the trend prediction skill of different similarity definitions and time window lengths p for the same K ($K = 10$, Fig. 3.4) in order to see if there is one best time window length for one K .

Fig. 3.4 (a) compares the skill for the different time window lengths p used in previous studies (Menary *et al.* 2021, Mahmood *et al.* 2022) with the same number of selected analogues ($K = 10$). Sensitivity to p appears highest for the 2D- and 3D RMSE-based and slightly less significant for the 2D correlation-based definition. The 2D RMSE-based definition has two time window length areas of significant skill separated by an area of no correlation around 20 years (compare Fig. 3.3 (a)). The stripes seen in Fig. 3.3 (b) show up for the 2D correlation-based definition, where time windows of length

3 Results

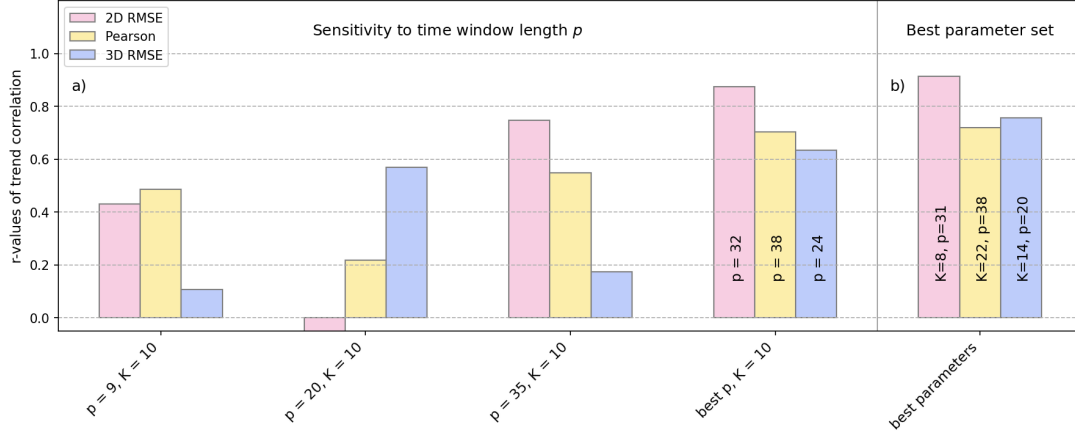


Fig. 3.4: Correlation of reconstructed trend time series with the reference (Rayner *et al.* 2003). Comparison of similarity definition (colored red, yellow and blue) for different parameter sets. a) shows the sensitivity to time window parameters from Mahmood *et al.* (2022), Menary *et al.* (2021) with the best parameters found for $K=10$; b) the best correlation values found for all parameters.

$p \approx 20$ years show no significant correlation. The best area for the 3D RMSE-based definition was found at $p \approx 20$ years in Fig. 3.3 (c) and this feature is visible in the comparison of $K = 10$.

The last three bars in Fig. 3.4 (a) show the time window lengths giving the highest skill for each similarity definition when selecting $K = 10$ analogues. The highest skill is found at $p = 32$ and 24 years for 2D and 3D RMSE, respectively, while $p = 38$ years yields the best skill for the 2D correlation-based definition, slightly increasing the values found with the window length from previous studies.

In Fig. 3.4 (b), we present the best parameter set for trend predictions across all K and p for each similarity definition. For the 2D RMSE-based definition, the highest skill is achieved with $K = 8, p = 31$, confirming the results from 3.2.1 that higher p yields better results (compare Fig. 3.3 (a)).

For the 2D correlation-based definition, the best parameter set is $K = 22, p = 38$. However, this result only slightly surpasses the highest skill found with $K = 10$ with the same time window length. This confirms that this similarity definition is less dependent on K and p .

Lastly, for the 3D RMSE-based definition, the best parameter set is $K = 14, p = 20$. The 3D RMSE-based definition proves to be the most sensitive to both parameters, and the best area in Fig. 3.3 (c) aligns with the parameters identified as optimal in Fig. 3.4. This suggests that there is not one best parameter set for high trend prediction skill across all similarity definitions, but that there are areas of highest skill and these areas are not the same across the similarity definitions. We want to look at the areas of optimal parameter sets in the next section.

3.3 Optimal parameter choice for skillful trend predictions

The analysis of the analogue method shows there is not one best parameter set for the prediction of trends, but each similarity definition has areas where skill is highest. We want to identify these areas by looking at the parameter sets above the 95% significance level (Fig. 3.5 (a)). For this purpose, we do not distinguish between similarity definitions and mark the parameter sets where one or more definitions have significant skill. We can see that almost all parameter sets can, in principle, be used to produce significantly skillful trend predictions, except smaller analogue ensembles $K < 6$ when time window length is $p < 30$. However, for ensemble sizes above or equal to 10 analogues, we see that there is significant skill for all time window lengths and a similarity definition with significant trend skill is consistently identifiable. Moreover, for many parameter sets, more than one similarity definition is applicable, indicating flexibility in the choice of the number of selected analogues and time window length.

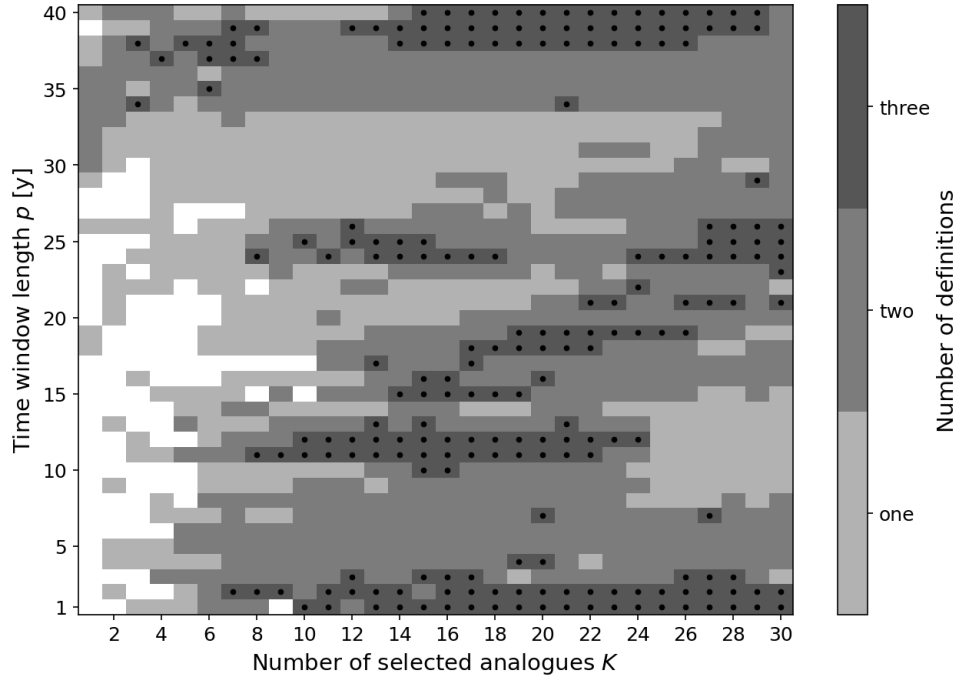
Another question is the optimal choice of similarity definition. We can see in Fig. 3.5 (a) that almost all parameter sets can be chosen, but can we also choose arbitrary similarity definitions for each of these? Since only a few parameter set areas are darkest grey with stipples (all three similarity definitions produce significant skill), this is not possible. We now look at the similarity definition producing the highest trend prediction skill for each parameter set (Fig. 3.5 (b)). We can see that the 2D RMSE-based definition and the 2D correlation-based definition outperform the 3D RMSE-based definition in most areas. Nevertheless, the 3D RMSE-based definition demonstrates superior skill in one major and two minor areas, suggesting that its inclusion can contribute to added overall skill of the analogue method.

For each parameter set there exists a best-performing similarity definition, but we can also choose a parameter set where all similarity definitions perform well. A choice can be made in a parameter area of stipple density, indicating robustness for all three similarity definitions. This can be found, for example, for $K = 15$ at $p = 11$ years or 38 years, optimizing the choice of time window length suggested by previous studies for the application to trend predictions (Mahmood *et al.* 2022, Menary *et al.* 2021).

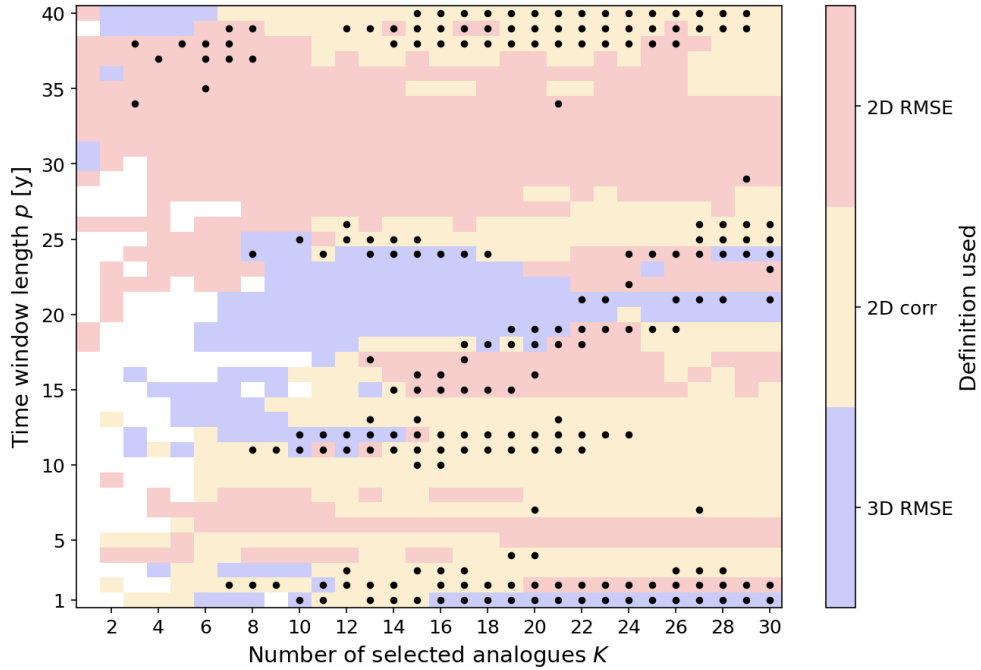
With this, we come to the end of the sensitivity analysis. This thesis aims at implementing the analogue method for trend predictions. First, we successfully implemented the analogue method for the MPI CMIP6 grand ensemble (Olonscheck *et al.* 2023) for which it hadn't been applied to before (see Section 3.1). Afterwards, we conducted a sensitivity analysis to the main choices of the application. We find that the analogue method is robust for trend predictions at $K > 10$, particularly for the two established similarity definitions, 2D RMSE and 2D correlation. While the 3D RMSE-based approach exhibits a slight reduction in overall skill compared to the 2D RMSE,

3 Results

it remains viable for trend reconstruction because of added skill in one area. We want to look at the predictions made by the analogue method next.



(a) Overview of parameter areas where one (light grey) or more (overlaps) similarity definitions exceed their respective 95% significance levels.



(b) Overview of the similarity definition with the highest skill of significant value.

Fig. 3.5: Areas of parameters with significant skill for all three similarity definitions. Stipples indicate where all three similarity definitions are above their 95% significance level simultaneously.

3.4 Trend predictions until 2029

In this section, we show the trend time series reconstructions for different parameter sets and look at the predictions made until 2029. We analyzed the analogue method on its sensitivity to various parameters and found that it is robust for the application to trend predictions. We found that it is important to choose a number of $K > 10$ different analogues to expect robust results.

Now we make trend predictions for the next ten years (2020–2029) with the analogue method using the MPI CMIP6 grand ensemble for the analogue selection. To give a measure of prediction uncertainty, we show prediction ensembles created with three similarity definitions. The choice of parameters is determined with the previous analysis.

3.4.1 Prediction with best overall skill

We first choose the parameter set that produces the highest trend prediction skill for each similarity definition. Fig. 3.6 shows the trend reconstruction and prediction created with these respective parameter sets. We can see a very close resemblance of the reference with the reconstruction in the years without volcanic influence (highlighted by grey lines). Especially the 2D RMSE-based definition reconstructs the observed trend time series very well, with a correlation of 0.91. The 2D correlation-based and

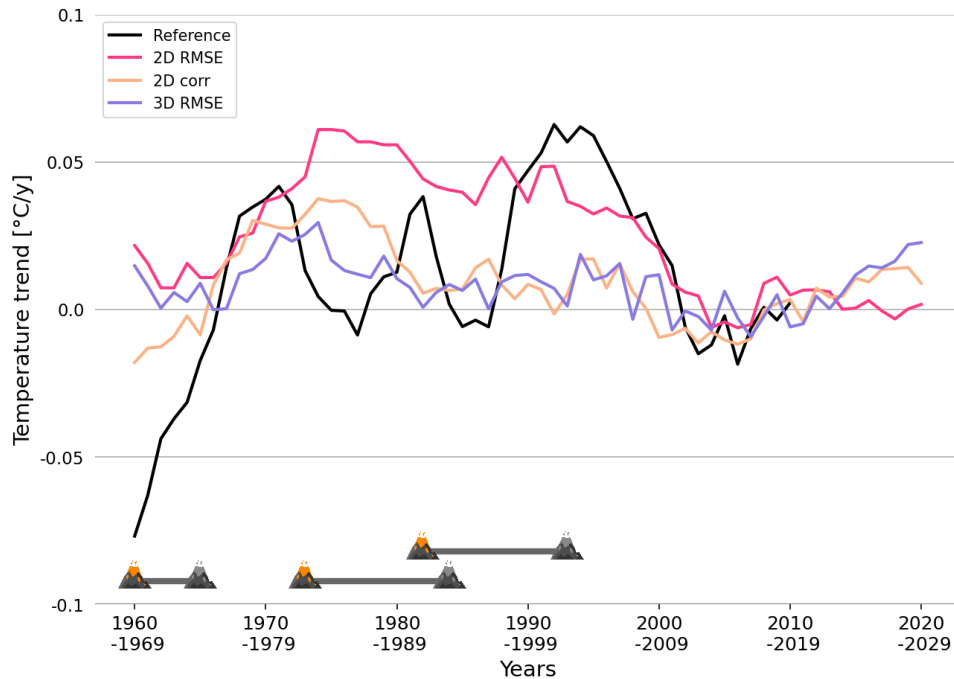


Fig. 3.6: Trend time series reconstruction for the best parameter set for each similarity definition. Marked areas are not considered for correlation calculation because they include volcanic eruptions. 2D RMSE-based (red): $K = 8, p = 31$, 2D correlation-based (yellow): $K = 22, p = 38$, 3D RMSE-based (purple): $K = 14, p = 20$. Reference in black (Rayner *et al.* 2003).

the 3D RMSE-based definition are slightly less skillful, with correlations of 0.72 and 0.76, respectively. However, these reconstructions provide a highly skillful prediction ensemble.

The ten-year intervals influenced by the volcanoes show a large difference between 2D RMSE and the other two definitions. All of them fail to predict the cooling in the early 1960s, the trend decrease in the mid 1970s and the trend increase at the end of the 19th century. This emphasizes the significance of the influence of volcanic eruptions on the North Atlantic SST and why it is necessary to exclude this influence from the estimation of trend prediction skill for a future where we have to assume no volcanic influence. In the intervals considered for the correlation, the trends resemble the reference closely, except for 2D correlation-based and 3D RMSE-based definition in the intervals at the end of the 1990s and the early 2000s.

The prediction of the 2020–2029 interval differs depending on the similarity definition, but all of them predict positive trends (2D RMSE-based: $0.002\text{ }^{\circ}\text{C}$ per year, 2D correlation-based: $0.009\text{ }^{\circ}\text{C}$ per year, 3D RMSE-based: $0.023\text{ }^{\circ}\text{C}$ per year), with a mean of $0.011\text{ }^{\circ}\text{C}$ per year $\pm 0.009\text{ }^{\circ}\text{C}$ per year. This corresponds to an expected mean warming in the North Atlantic of about $0.11\text{ }^{\circ}\text{C} \pm 0.09\text{ }^{\circ}\text{C}$ until 2029. The uncertainty range shows that a warming (positive trend) is expected for the 2020s.

3.4.2 Prediction with robust parameter sets

We now want to compare our choice from Section 3.3, the best parameter set for each similarity definition selected individually, with robust parameter sets, i.e., a parameter selection where all three similarity definitions show significant trend correlations. In Section 3.3 we found that the choice of K is less important, as long as it is chosen to be above or equal to 10. We choose $K = 15$ here, because it is an intermediate choice between too many (which showed smaller skill in Fig. 3.3) and too few (no significant skill) selected analogues. The suggested choice of time window lengths in Section 3.3 was 11 years and 38 years, being in areas of denser stipples, and we use these lengths for the prediction made by robust parameter set.

In Fig. 3.7 we show the results of the two parameter sets chosen from Fig. 3.5 (a). Fig. 3.7 (a) depicts the trend time series reconstruction for $K = 15$ and $p = 11$ years. The correlation of the time mean-based definitions is 0.51 for both, the 3D RMSE-based definitions shows 0.47. Fig. 3.7 (b) shows the results for $K = 15$ and $p = 38$ years. The correlation of the time mean-based definitions is 0.60, the 3D RMSE-based definitions shows 0.41.

Both sets predict a mean trend for 2020–2029 of $0.005\text{ }^{\circ}\text{C}$ per year $\pm 0.007\text{ }^{\circ}\text{C}$ per year. This is smaller than for the best parameters sets and has a larger standard deviation. We expect a warming of the North Atlantic by $0.05\text{ }^{\circ}\text{C} \pm 0.07\text{ }^{\circ}\text{C}$ until 2029 with this approach.

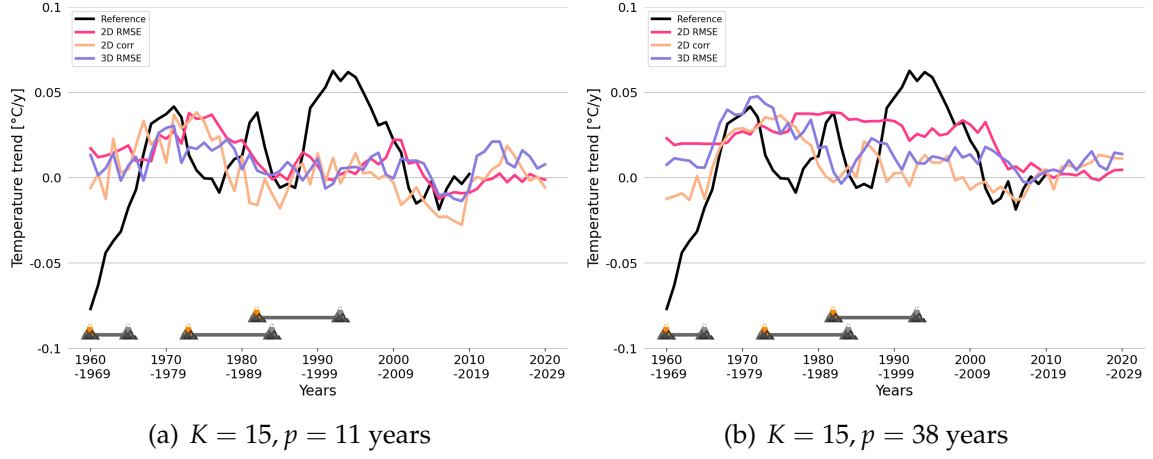


Fig. 3.7: Trend time series reconstruction with two sets of parameters where all similarity definition showed significant trend skill. Marked areas are not considered for correlation calculation because they include volcanic eruptions. The similarity definitions are shown as: 2D RMSE-based (red), 2D correlation-based (yellow), 3D RMSE-based (purple). Reference in black (Rayner *et al.* 2003).

When comparing the spread of predictions for 2020–2029 in Fig. 3.7 with Fig. 3.6, we see a larger spread in Fig. 3.7 relative to the prediction. While the absolute spread of the prediction in Fig. 3.6 was larger, it was not larger than the mean prediction itself. The choice of a robust parameter set seems to lead to larger spread of the predictions. Considering the lower reconstruction correlation, we would rather opt for a combination of different best parameter sets, leading to a larger ensemble of predictions that can be viewed in terms of mean and variation.

4 Discussion

This thesis aimed at predicting SST trends with the analogue method using the MPI CMIP6 grand ensemble. For this purpose, we started with a systematic analysis of sensitivity to the choices of parameters and created predictions with the optimal parameters found in this analysis. Our results indicate general robustness of the analogue method for the application to trend predictions with dependence on the similarity definition. The analysis revealed that volcanic eruptions significantly impact the skill of trend predictions. We now assess the literature for a comparison and discussion of our results.

We started our study with the implementation of the analogue method using the MPI CMIP6 grand ensemble instead of the large CMIP archives. For the first time, we applied the analogue method to decadal trend predictions. There are a few choices to be made when implementing the method and we started with the parameters suggested by previous studies (Mahmood *et al.* 2022, Menary *et al.* 2021). We were able to provide a reconstruction of the trend time series from 1960 to 2019 with high correlation with the reference of observed trends in the same time period.

Continuing, we compared the influence of volcanic eruptions on the trend skill. Previous studies had already mentioned that the analogue method is not equipped to predict volcanoes and the corresponding upper ocean cooling (Mahmood *et al.* 2021, Menary *et al.* 2021), but they did not account for this limitation. It is believed that the influence of volcanoes provides an uncertainty on decadal prediction skill assessments and that external forcings are a source of skill in the North Atlantic (Yeager and Robson 2017). In the frame of this thesis, we could not provide an analysis of the influence of external forcings like volcanoes but we compared the prediction skill of our trend time series reconstruction with and without volcano years.

We found that trend time series including years of major volcanic eruptions tend to overestimate the importance of a long-term trend in the trend time series. The correlation of time series with volcano years was higher when an overall long-term trend was met than when the individual ten-year trends were highly correlated. Concluding, we found that time series with negative trends around volcano years were valued higher by the correlation than time series with a good reconstruction of the early 2000s that did not have these negative trends.

This suggests a bias to the volcanic cooling effects. The analogue method is not designed to predict volcanoes but rather the North Atlantic SPG patterns and SST in general. By excluding the volcano years from the correlation calculation, we tried to emphasize this feature and evaluate the analogue method's ability to predict it.

The influence of volcanoes has been subject to discussion for the past years and this work adds to the question: How much does the knowledge we have at a current date (in this case about volcanic eruptions) influence the skill of reconstructions of the past? This is important for making skill estimation that are assessing the ability of a method or model to predict the future, where no such prior knowledge is available. Our findings suggest the necessity to compare simulations where no prior knowledge about volcanic eruptions has been applied, in order to analyze this feature.

For this work, we were able to focus the trend reconstruction mainly on the time period after 1993 where no volcanoes influenced the North Atlantic to the present day. This provides a long time period without truncation for prediction skill assessments. Additionally, the analysis presented in this thesis is a good starting point for investigating the volcanic influence on trend prediction skills. While we initially followed the approach of previous studies with the reconstruction of 1960–2019, future studies could focus their research entirely on the time period after 1993.

The analogue method needs a well observed variable for pattern comparison and we chose North Atlantic SST because of oceanic memory. Prediction skill is expected to be highest in the region of North Atlantic SPG as this is the dominant pattern on decadal time scales. In Fig. 3.2 we were able to confirm this assumption by correlating the reconstructed trends with the trends of each grid cell in the reference.

Previous studies have focused on lead years instead of trend predictions on the same decadal time scales. Mahmood *et al.* (2022) show a map of anomaly correlation of the period 1962 until 2019 with less skill in the North Atlantic SPG region than in other areas, yet the skill in this region is comparable to the skill our approach exhibited (4.1 (a)). Menary *et al.* (2021) reconstructed the time series from 1960 to 2019 of lead years 2–10 combined. Their skill for the SPG compares well with the skill of our approach (see Fig. 4.1 (b)).

Concluding, the application of the analogue method for trend reconstruction is utilizing the information provided by dominant SST patterns to determine analogues in much the same way that previous studies have revealed.

To assess the sensitivity of the analogue method we performed an in-depth analysis of the major choices of parameters. We found that this application of the analogue method is robust for trend predictions when using time-mean based similarity definitions and shows good results with large robust areas for the new time-dependent definition.

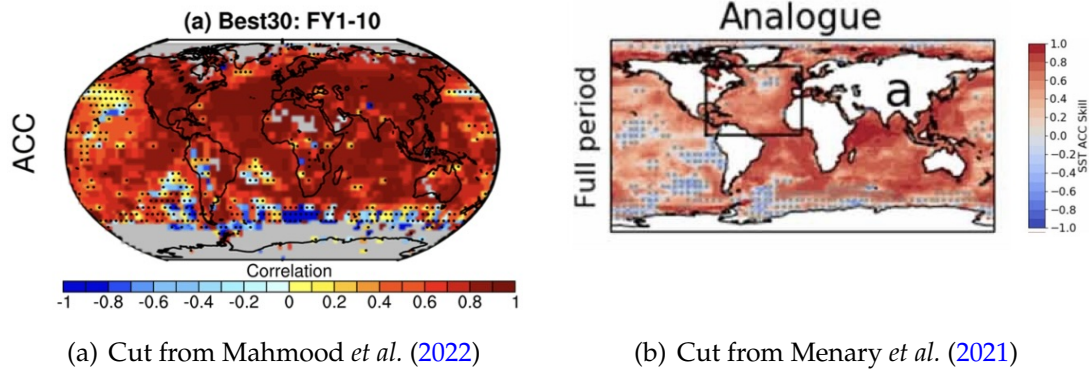


Fig. 4.1: Map of anomaly correlation coefficient from Mahmood *et al.* (2022) and Menary *et al.* (2021) for 1960–2019 of (a) lead year 1–10 combined, and (b) lead year 2–10 combined. For comparison, we concentrate only on the North Atlantic.

In Section 3.2.1 we compared the skill for different numbers of analogues selected per time step and found that above a certain threshold of approximately $K = 10$ the trend prediction skill becomes largely independent of the number of selected analogues. This offers new opportunities for the application of the analogue method. The analogue method selects the best matches independent of their assigned time stamp, leading to a re-combination with repetition. Our findings open up the possibility of creating an additional number of time series from a limited number of simulations. Even a single ensemble member can be used to create a prediction with more than one analogue ensemble member, when repetition for each time step is allowed.

When comparing different time window lengths, we found a larger dependence on the choice of this parameter. However, above the analogue ensemble size threshold of $K = 10$, the time-mean similarity definitions showed significant skill for almost all p .

The major part of the analysis focused on comparing different similarity definitions. Menary *et al.* (2021) and Mahmood *et al.* (2021) used only one similarity definition, assuming that the choice of the mathematical comparison procedure does not influence the results. Mahmood *et al.* (2022) mentioned that skill with a 2D RMSE-based definition fell behind the skill of the 2D correlation-based definition. All of the similarity definitions were based on a time mean state comparison of simulations and reference. To investigate the importance of time-dependent SST pattern evolution, we compared the established definitions with the time-dependent 3D RMSE-based definition. Our results confirm the robustness of the analogue method to the similarity definitions introduced by Mahmood *et al.* (2021) and Menary *et al.* (2021), where both time-mean definitions showed similar skill for large areas of the two main parameters. The 3D RMSE-based definition showed lower skill in many areas, but proved to give added skill in areas where the other two definitions came short. We were expecting different behavior of the time-dependent definition, especially for long time windows, because

of the long-term evolution of SST patterns. This showed in the overview of different (K, p) , where we found that the intermediate choice of time window lengths leads to the best results for this definition. The lower skill in the areas of longer time windows probably comes from the difficulty of finding good analogues for such a complex comparison. Each year is compared with this method, meaning for large p that a good analogue needs to match up over 25 or more years. Shorter time windows for this definition might be advisable.

Our findings suggest that all of these definitions can be used to predict temperature trends in the North Atlantic, and a combination of the definitions can be chosen depending on the application.

After the analysis of the analogue method we were able to make a prediction of the trend for the ten-year interval 2020–2029. We found that the temperature of the North Atlantic is expected to rise by $0.005 - 0.011\text{ }^{\circ}\text{C} \pm 0.007 - 0.009\text{ }^{\circ}\text{C}$ per year. Different similarity definitions differ in their prediction, with the standard deviation of a set of three predictions, one per similarity definition, being on the scale of prediction itself. This is why using a set of predictions is important and useful in reducing uncertainty. It may be advisable to increase the number of predictions for future studies, and the analysis of this thesis can serve as a starting point for the selection of parameters. Additionally, the prediction of the best parameter set showed less relative spread than the predictions of robust sets. Future research could focus on a prediction ensemble made from a number of best parameter sets, since higher reconstruction correlations lead to better predictions.

When we look at the temperatures measured in the North Atlantic for the last four years (2020–2023), we can compare our results for 2020–2029 with the real world in the first years. From 2020–2023, the mean temperature in our North Atlantic box has changed from $9.28\text{ }^{\circ}\text{C}$ in 2020 to $9.52\text{ }^{\circ}\text{C}$ in 2023, with a slope of $0.07\text{ }^{\circ}\text{C}$ per year. This lies in between the results of our prediction for 2020–2029, with $0.005\text{--}0.011\text{ }^{\circ}\text{C}$ per year. The trends are calculated for ten-year intervals, so a final assessment can only be complete at the end of the decade which is why we only included this short comparison.

In Menary *et al.* (2021), the area average SST of lead year 2–10 combined decreased for the years following 2020 which suggests a negative trend in the time interval 2020–2029. Given the recent mean SST evolution for 2020 to 2023, we can conclude that our approach was able to predict this evolution with higher accuracy. Mahmood *et al.* (2022) did not show a prediction for 2020–2029 in the North Atlantic.

The best analogue of a data set does not necessarily have to be a good analogue. The quality we can expect from an analogue depends on the length of the compared data sets (van den Dool 1994), which is why simulation data was chosen in this thesis. This leads to a data set large enough to expect good analogues. However, the choice of the

time window length influences the number of analogues available. Long time window lengths shorten the selectable analogues more than short time window lengths: The data is available from 1850 to 2100, the last time interval with a time window length of $p = 35$ years is 2056, while for $p = 5$ years it is 2085, leaving 30 more time intervals to choose from for every ensemble member. Overall, the time window length shortens the time series from 240 years total to $240 - p + 1$ years. Our results suggest that this influence is marginal because time windows after 2014 were rarely chosen as analogues. Further analysis could investigate how good the best matches (analogues) for the simulation based analogue method are.

When searching for the best analogue for each time step, one possibility is the selection of the historical simulation ensemble members for this time step. This means that when searching the analogue for the observations from 1960 to 1970 ($p = 11$ years), the analogue method could select the best match of the historical members representing 1960 to 1970. Yet, when we analyzed the time stamps of the selected historical members, we found that the same time window was rarely chosen.

The historical simulation is created from the piControl simulation by introducing time dependent external forcings, from 1850 onward and at every time step. These forcings are created based on what was observed for this specific time step in the past. This, however, cannot provide predictions for the future, since we cannot constrain the initial conditions with external forcings and the historical simulations cannot provide predictions. Usually, this is done by using external forcing and simultaneously phasing in observations of the real climate into the Earth system model ("assimilation"), with the aim to start multi-year "initialized" climate predictions (e.g. Brune and Baehr 2020). The analogue method provides a simplified constraining procedure by finding the best match of initial spatial SST patterns in the simulation. Because this approach does not select the historical members with the same time stamp as the reference in the past, we conclude that the spatial patterns of SST are not best simulated by the external forcings.

The sensitivity analysis indicates that it is possible to optimize the skill for specific applications by choosing the parameters and similarity definitions leading to highest skill. In this thesis, we selected the parameters with highest skill in the trend reconstruction to provide a prediction for 2020-2029. The same can be done to create lead year predictions commonly used in the decadal prediction community (see A1. Supplements: commonly used lead2-5 combined, Boer *et al.* 2016). In this work, we did not focus on individual lead year reconstruction, but Fig. S1-S3 suggest high skill for this application with the simulation data set provided by the MPI CMIP6 grand ensemble. This can be interesting for future analysis of the analogue method using this simulation ensemble.

5 Summary and Conclusion

In this thesis, we applied the analogue method to SST trend predictions of the North Atlantic. Using the MPI CMIP6 grand ensemble we selected analogues of observed SST patterns in order to provide a simple and fast prediction scheme using existing data sets. We summarize the main findings:

- We extended the application of the analogue method to predict temperature trends. **Our analysis indicates that the analogue method can be applied for skillful SST trend predictions in the North Atlantic, with prediction skill up to 0.9. Skill was found in the regions of major SST patterns in the North Atlantic, created by the North Atlantic subpolar gyre (SPG).**
- **Based on a sensitivity analysis of the main choices for parameters, we can confirm overall robustness of the analogue method.** This offers a wide variety of future applications of the analogue method.
- We introduced a new similarity definition to compare the influence of SST pattern time evolution. **This similarity definition leads to added skill for parameters where the time mean similarity definitions fell short.** Additionally, further research suggests that this new similarity definition could be useful for lead year predictions.
- We compared the prediction skill of a trend time series with volcanic eruptions to the trend time series without them. **This suggests that volcanic-driven upper ocean cooling dominates the trend so that the overall correlation is higher when this cooling is represented than when other parts of the time series are highly correlated. This bias needs to be addressed when providing skill assessments in time periods with major volcanic eruptions.**
- An ensemble of predictions can be used for predictions with uncertainty spread. We suggest an ensemble made from parameter sets and similarity definitions with highest prediction skill. **Our results indicate an anticipated warming of the North Atlantic sea surface until 2029.**

In conclusion, we were able to extend the application of the analogue method to temperature trend prediction with the MPI CMIP6 grand ensemble. We discussed new

5 *Summary and Conclusion*

possibilities of applications, ranging from re-selection schemes over historical member analysis to lead year analysis using time evolution of SST patterns with a new similarity definition. Our results suggest the need for research on the influence of volcanic eruptions on the prediction skill, adding to the ongoing discussion of the matter.

A1 Supplements

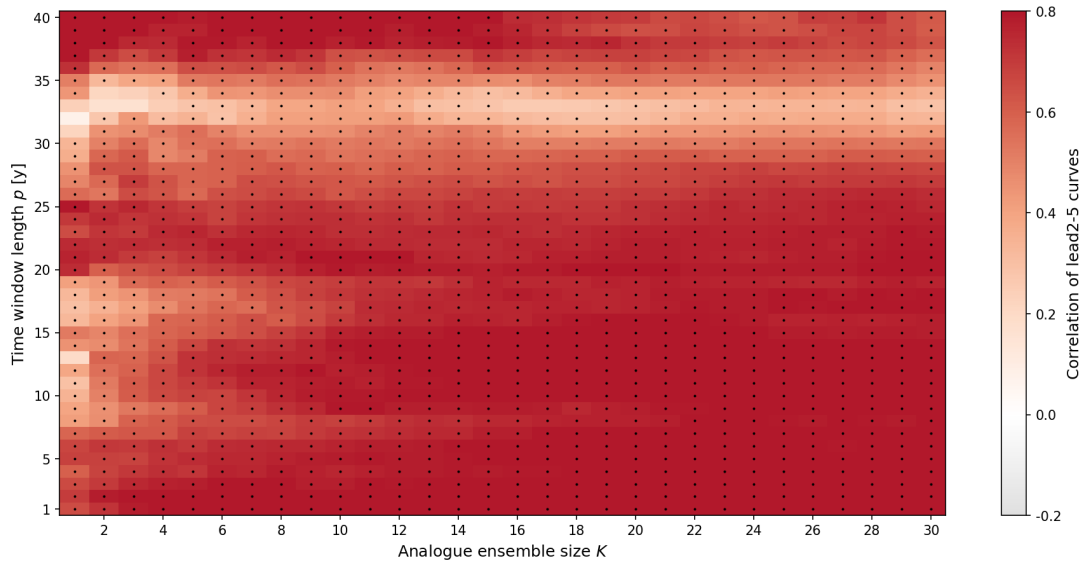


Fig. S1: Skill of reconstructing lead year 2–5 combined for 2D RMSE-based definition. Volcano years are not yet excluded in this plot. Stipples indicate significant correlation.

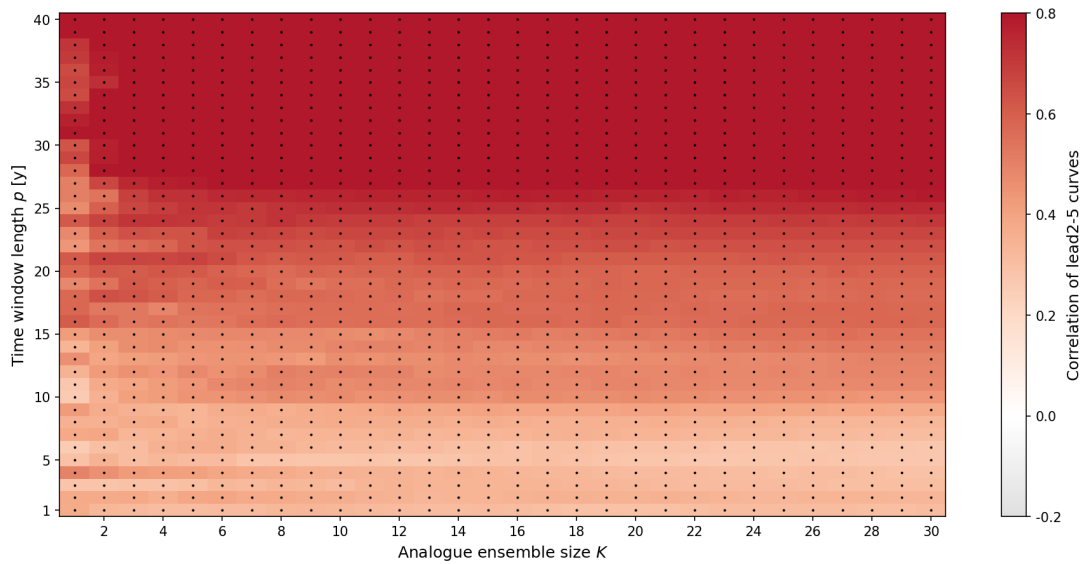


Fig. S2: Skill of reconstructing lead year 2–5 combined for 2D correlation-based definition. Volcano years are not yet excluded in this plot. Stipples indicate significant correlation.

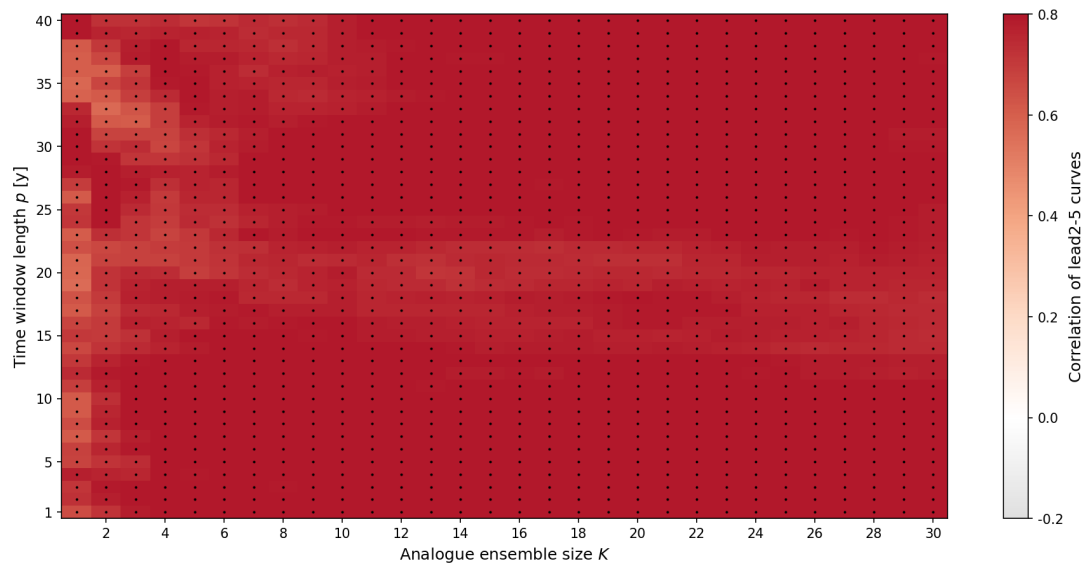


Fig. S3: Skill of reconstructing lead year 2–5 combined for 3D RMSE-based definition. Volcano years are not yet excluded in this plot. Stipples indicate significant correlation.

Literature

- Boer, G. J., D. M. Smith, C. Cassou, F. Doblas-Reyes, G. Danabasoglu, B. Kirtman, Y. Kushnir, M. Kimoto, G. A. Meehl, R. Msadek, *et al.* (2016). “The decadal climate prediction project (DCPP) contribution to CMIP6”. In: *Geoscientific Model Development* 9.10, pp. 3751–3777.
- Borchert, L. F., W. A. Müller, and J. Baehr (2018). “Atlantic ocean heat transport influences interannual-to-decadal surface temperature predictability in the North Atlantic region”. In: *Journal of Climate* 31.17, pp. 6763–6782.
- Borchert, L. F., H. Pohlmann, J. Baehr, N.-C. Neddermann, L. Suarez-Gutierrez, and W. A. Müller (2019). “Decadal predictions of the probability of occurrence for warm summer temperature extremes”. In: *Geophysical Research Letters* 46.23, pp. 14042–14051.
- Born, A., J. Mignot, and T. F. Stocker (2015). “Multiple equilibria as a possible mechanism for decadal variability in the North Atlantic Ocean”. In: *Journal of Climate* 28.22, pp. 8907–8922.
- Brown, P. T. and K. Caldeira (2020). “Empirical prediction of short-term annual global temperature variability”. In: *Earth and Space Science* 7.6, e2020EA001116.
- Brune, S. and J. Baehr (2020). “Preserving the coupled atmosphere–ocean feedback in initializations of decadal climate predictions”. In: *Wiley Interdisciplinary Reviews: Climate Change* 11.3, e637.
- Buckley, M. W., T. DelSole, M. S. Lozier, and L. Li (2019). “Predictability of North Atlantic sea surface temperature and upper-ocean heat content”. In: *Journal of Climate* 32.10, pp. 3005–3023.
- Ding, H., M. Newman, M. A. Alexander, and A. T. Wittenberg (2019). “Diagnosing secular variations in retrospective ENSO seasonal forecast skill using CMIP5 model-analogs”. In: *Geophysical Research Letters* 46.3, pp. 1721–1730.
- Driscoll, S., A. Bozzo, L. J. Gray, A. Robock, and G. Stenchikov (2012). “Coupled Model Intercomparison Project 5 (CMIP5) simulations of climate following volcanic eruptions”. In: *Journal of Geophysical Research: Atmospheres* 117.D17.

- Eyring, V., S. Bony, G. A. Meehl, C. A. Senior, B. Stevens, R. J. Stouffer, and K. E. Taylor (2016). “Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6) experimental design and organization”. In: *Geoscientific Model Development* 9.5, pp. 1937–1958.
- Griffies, S. M. and K. Bryan (1997). “Predictability of North Atlantic multidecadal climate variability”. In: *Science* 275.5297, pp. 181–184.
- Hawkins, E., J. Robson, R. Sutton, D. Smith, and N. Keenlyside (2011). “Evaluating the potential for statistical decadal predictions of sea surface temperatures with a perfect model approach”. In: *Climate Dynamics* 37, pp. 2495–2509.
- Hermanson, L., R. Eade, N. H. Robinson, N. J. Dunstone, M. B. Andrews, J. R. Knight, A. A. Scaife, and D. M. Smith (2014). “Forecast cooling of the Atlantic subpolar gyre and associated impacts”. In: *Geophysical Research Letters* 41.14, pp. 5167–5174.
- Jungclaus, J. H., N. Fischer, H. Haak, K. Lohmann, J. Marotzke, D. Matei, U. Mikolajewicz, D. Notz, and J. von Storch (2013). “Characteristics of the ocean simulations in the Max Planck Institute Ocean Model (MPIOM) the ocean component of the MPI-Earth system model”. In: *Journal of Advances in Modeling Earth Systems* 5.2, pp. 422–446.
- Kriegler, E., B. C. O’Neill, S. Hallegatte, T. Kram, R. J. Lempert, R. H. Moss, and T. Wilbanks (2012). “The need for and use of socio-economic scenarios for climate change analysis: a new approach based on shared socio-economic pathways”. In: *Global Environmental Change* 22.4, pp. 807–822.
- Lorenz, E. N. (1969). “Atmospheric predictability as revealed by naturally occurring analogues”. In: *Journal of Atmospheric Sciences* 26.4, pp. 636–646.
- Mahmood, R., M. G. Donat, P. Ortega, F. J. Doblas-Reyes, C. Delgado-Torres, M. Samsó, and P.-A. Bretonnière (2022). “Constraining low-frequency variability in climate projections to predict climate on decadal to multi-decadal timescales—a poor man’s initialized prediction system”. In: *Earth System Dynamics* 13.4, pp. 1437–1450.
- Mahmood, R., M. G. Donat, P. Ortega, F. J. Doblas-Reyes, and Y. Ruprich-Robert (2021). “Constraining Decadal Variability Yields Skillful Projections of Near-Term Climate Change”. In: *Geophysical Research Letters* 48.24, e2021GL094915.
- Marsland, S. J., H. Haak, J. H. Jungclaus, M. Latif, and F. Röske (2003). “The Max-Planck-Institute global ocean/sea ice model with orthogonal curvilinear coordinates”. In: *Ocean Modelling* 5.2, pp. 91–127.

- Mauritsen, T., J. Bader, T. Becker, J. Behrens, M. Bittner, R. Brokopf, V. Brovkin, M. Claussen, T. Crueger, M. Esch, *et al.* (2019). “Developments in the MPI-M Earth System Model version 1.2 (MPI-ESM1.2) and its response to increasing CO₂”. In: *Journal of Advances in Modeling Earth Systems* 11.4, pp. 998–1038.
- Menary, M. B., J. Mignot, and J. Robson (2021). “Skilful decadal predictions of subpolar North Atlantic SSTs using CMIP model-analogues”. In: *Environmental Research Letters* 16.6, p. 064090.
- Monerie, P.-A., J. Robson, B. Dong, and N. Dunstone (2018). “A role of the Atlantic Ocean in predicting summer surface air temperature over North East Asia?” In: *Climate Dynamics* 51, pp. 473–491.
- O’Neill, B. C., C. Tebaldi, D. P. van Vuuren, V. Eyring, P. Friedlingstein, G. Hurtt, R. Knutti, E. Kriegler, J.-F. Lamarque, J. Lowe, *et al.* (2016). “The scenario model intercomparison project (ScenarioMIP) for CMIP6”. In: *Geoscientific Model Development* 9.9, pp. 3461–3482.
- O’Neill, B. C., E. Kriegler, K. Riahi, K. L. Ebi, S. Hallegatte, T. R. Carter, R. Mathur, and D. P. van Vuuren (2014). “A new scenario framework for climate change research: the concept of shared socioeconomic pathways”. In: *Climatic Change* 122, pp. 387–400.
- Olonscheck, D., L. Suarez-Gutierrez, S. Milinski, G. Beobide-Arsuaga, J. Baehr, F. Fröb, T. Ilyina, C. Kadow, D. Krieger, H. Li, J. Marotzke, É. Plésiat, M. Schupfner, F. Wachsman, L. Wallberg, K.-H. Wieners, and S. Brune (2023). “The New Max Planck Institute Grand Ensemble With CMIP6 Forcing and High-Frequency Model Output”. In: *Journal of Advances in Modeling Earth Systems* 15.10, e2023MS003790.
- Pohlmann, H., J. H. Jungclaus, A. Köhl, D. Stammer, and J. Marotzke (2009). “Initializing decadal climate predictions with the GECCO oceanic synthesis: Effects on the North Atlantic”. In: *Journal of Climate* 22.14, pp. 3926–3938.
- Rayner, N., D. E. Parker, E. Horton, C. K. Folland, L. V. Alexander, D. Rowell, E. C. Kent, and A. Kaplan (2003). “Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century”. In: *Journal of Geophysical Research: Atmospheres* 108.D14.
- Robock, A. (2000). “Volcanic eruptions and climate”. In: *Reviews of Geophysics* 38.2, pp. 191–219.
- Robock, A. and J. Mao (1995). “The volcanic signal in surface temperature observations”. In: *Journal of Climate* 8.5, pp. 1086–1103.

- Schulzweida, U. (2023). *CDO User Guide*. Version 2.3.0. URL: <https://doi.org/10.5281/zenodo.10020800>.
- Smith, D. M., R. Eade, and H. Pohlmann (2013). "A comparison of full-field and anomaly initialization for seasonal to decadal climate prediction". In: *Climate Dynamics* 41, pp. 3325–3338.
- Sutton, R. T. and D. L. Hodson (2005). "Atlantic Ocean forcing of North American and European summer climate". In: *Science* 309.5731, pp. 115–118.
- Swingedouw, D., P. Ortega, J. Mignot, E. Guilyardi, V. Masson-Delmotte, P. G. Butler, M. Khodri, and R. Séférian (2015). "Bidecadal North Atlantic ocean circulation variability controlled by timing of volcanic eruptions". In: *Nature Communications* 6.1, p. 6545.
- Talley, L. D. (2011). *Descriptive physical oceanography: an introduction*. Academic press.
- Timmreck, C., H. Pohlmann, S. Illing, and C. Kadow (2016). "The impact of stratospheric volcanic aerosol on decadal-scale climate predictions". In: *Geophysical Research Letters* 43.2, pp. 834–842.
- van den Dool, H. (1994). "Searching for analogues, how long must we wait?" In: *Tellus A* 46.3, pp. 314–324.
- Wieners, K.-H., M. Giorgetta, J. Jungclaus, C. Reick, M. Esch, M. Bittner, V. Gayler, H. Haak, P. de Vrese, T. Raddatz, T. Mauritsen, J.-S. von Storch, J. Behrens, V. Brovkin, M. Claussen, T. Crueger, I. Fast, S. Fiedler, S. Hagemann, C. Hohenegger, T. Jahns, S. Kloster, S. Kinne, G. Lasslop, L. Kornblueh, J. Marotzke, D. Matei, K. Meraner, U. Mikolajewicz, K. Modali, W. Müller, J. Nabel, D. Notz, K. Peters-von Gehlen, R. Pincus, H. Pohlmann, J. Pongratz, S. Rast, H. Schmidt, R. Schnur, U. Schulzweida, K. Six, B. Stevens, A. Voigt, and E. Roeckner (2023). *IPCC DDC: MPI-M MPIESM1.2-LR model output prepared for CMIP6 ScenarioMIP*. URL: <https://doi.org/10.26050/WDCC/AR6.C6SPMXML2>.
- Wilks, D. S. (2011). *Statistical methods in the atmospheric sciences*. Vol. 100. Academic press.
- Yeager, S. G. and J. Robson (2017). "Recent progress in understanding and predicting Atlantic decadal climate variability". In: *Current Climate Change Reports* 3, pp. 112–127.

- Zorita, E., J. P. Hughes, D. P. Lettemaier, and H. von Storch (1995). "Stochastic characterization of regional circulation patterns for climate model diagnosis and estimation of local precipitation". In: *Journal of Climate* 8.5, pp. 1023–1042.
- Zorita, E. and H. von Storch (1999). "The analog method as a simple statistical down-scaling technique: Comparison with more complicated methods". In: *Journal of Climate* 12.8, pp. 2474–2489.

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